# MODULE 8 TOYOTA ANALYSIS

# Toyota

(includes Lexus and Scion)

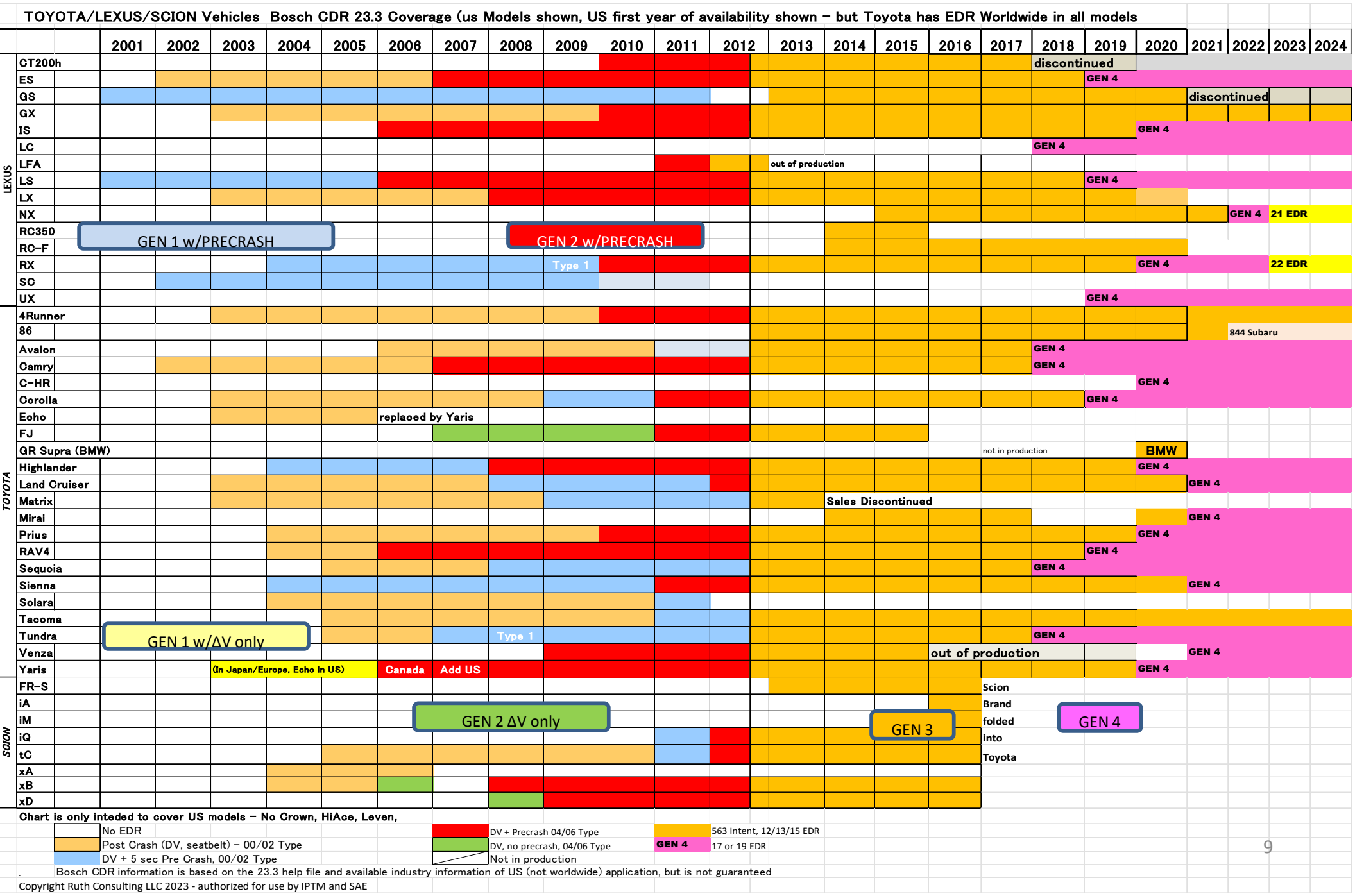
- Toyota began phasing in EDR's as early as 2000. The first model with EDR, the Lexus GS, had both Delta V and pre-crash data.
- Some 2001-2010 models have only longitudinal Delta V. By 2011 ALL have pre-crash data.
- All Toyotas can record rear impacts.
- Gen 1 & 2 recording threshold is wake-up.
- In addition to regular interval precrash data, Toyotas record an ADDITIONAL data point at or near algorithm enable, which Toyota calls TRG (trigger).

# Toyota

(includes Lexus and Scion)

- Records frontal data in frontal algo wakeups/events, side data in side wakeups/events if equipped, from their respective AE's. Front & Side NOT on same time zero.
- Side data has shorter time than frontal data until 13EDR
- Rollover events if equipped, from rollover algo AE.
- Other manufacturers have all rules in the Data Lims.
- Some Toyota definitions are in CDR "Glossary".
- "**RTFG**" – **R**ead **T**he **F**riggin' **G**lossary"

# Toyota CDR 23.3 Coverage US Models



# Toyota Unique Help File Organization

## Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in so
note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Click here for a list of s

Click here for tips on how to use this *Supported Vehicles* help topic

**IMPORTANT:** To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the ap

**General Toyota Application Notes:**

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will n
vehicles listed in the help file below are supported as indicated by the list and any associated coverage notes.

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287)

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vel

**Vehicle Specific Application Notes:**

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W Y Z

**4Runner (Toyota)** **2003 and newer** **Market: 7**

| Year | Module | | Vehicle Interface | OBD/DLC Connect Adapter/Cable | D2M Connect Adapter/Cable | Module Location |
|---|---|---|---|---|---|---|
| 2002 - 2008 | ACM | Data - 02EDR | CANplus or CDR 900 | CANplus - F00K108287 - *for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to* | CANplus - (no adapter) Cable#F00K108617 CDR 900 - click here (use listed cable above and no ACM adapter) | Under Center Stack |

# New Toyota Vehicle Lookup

**Camry (Toyota)** **2001 and newer** **Market: 7**

7=All Markets the Vehicle is Sold

| Year | Module | | Vehicle Interface | OBD/DLC Connect Adapter/Cable | D2M Connect Adapter/Cable | Module Location |
|---|---|---|---|---|---|---|
| 2001 - 2004 (all markets)<br><br>2005 (US/Canada) | ACM | Data - 00EDR / 02EDR | CANplus<br>or<br>CDR 900 | CANplus - F00K108287<br>*- for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the US &*<br>CDR 900 - click here | CANplus - (no adapter) Cable#F00K108615<br>- (no adapter) Cable#F00K108617<br>Click here [?] to match the ACM part number to the correct cable<br>CDR 900 - click here (use listed cable above and no ACM adapter) | Under Center Stack |

Multiple Cables for same model

Must Look up Cable for DTM by Part No

# **N**ew Toyota Help File ***U.S. Issues***

- Because the help file applies WORLDWIDE, it lists MANY models not found in the US/CAN.
- Toyotas internal records may be by ***CALENDAR*** YEAR, not ***MODEL*** YEAR. Help File may show EDR is first available in 2003, but that was 2004 Model Year in the US. MOST models are “one year off”.
- Avalon shows EDR in 02 but prior charts showed no EDR until 06 – do not assume EDR present in every vehicle listed in the help file

# Unique to Toyota

- Delta V is recorded real time as it is occurring (GM waits until event is over to write anything). Even in older models with no backup power supply for recording, partially written Delta V is possible
- There are multiple “recording complete” entries:
- 1. Complete file
- 2. Crash Pulse subsection (Delta V), per event
- 3. Precrash subsection (Speed), per event

# Toyota EDR Evolution

| GEN 1<br>00,02 EDR | GEN 2<br>04,06 EDR | GEN 3<br>12,13,15 EDR | GEN 4<br>17,19 EDR |
|---|---|---|---|
| 3 front<br>1 side<br>1 roll | 2 front<br>2 side<br>2 roll<br>2 precrash | | 4 front<br>4 side<br>2 roll<br>6 precrash |
| Record on 2g wakeup | | Record on<br>5mph X$\Delta v$,<br>side 3.5g wake | Record on<br>5 X or Y<br>$\Delta v$ |
| 150ms X$\Delta v$<br>5PC@1 + TRG | 200ms X$\Delta v$<br>5PC@1+TRG | 250ms X$\Delta v$<br>10PC@0.5 + TRG | 250ms XY$\Delta v$<br>10PC@0.5+TRG |

# Toyota EDR Evolution cont'd

| GEN 1<br>00,02 EDR | GEN 2<br>04,06 EDR | GEN 3<br>12,13,15 EDR | GEN 4<br>17,19 EDR |
|---|---|---|---|
| 70ms YΔ$v$@ACM+sats | | 12 YDV 70ms<br>sats only<br>13 250ms ACM | 250ms<br>YDV |
| REFRESH<br>500 MS SPEED | | REFRESH<br>24 MS SPEED | |
| 7**8.3** MPH<br>TOP, 2kph res | 7**5.8** MPH TOP.<br>2kph res | 124.3 MPH/200kph TOP<br>1 kph res | |
| Accel Pedal Volts | | Accel Pedal % & Throttle % | |

# Toyota EDR Evolution cont'd

| GEN 1<br>00,02 EDR | GEN 2<br>04,06 EDR | GEN 3<br>12,13,15 EDR | GEN 4<br>17,19 EDR |
|---|---|---|---|
| RPM to next lower 400<br>Referresh rate 24ms | | RPM to 100 | |
| No Key Cycles | | Key Cycles | |
| Time Between Events < 5 sec | Time Between Events <16 sec | | |
| Brake on registers immediately<br>No delay in reporting | | Brk Oil Pres, Stab. Cont. Long Accel | |
| | | Strg & Yaw | |

# 00/02 EDR record summary

**Front/Rear Event Record Summary at Retrieval**

Side Events not numbered

| Events Recorded | TRG Count | Crash Type | Time (msec) | Event & Crash Pulse Data Recording Status |
|---|---|---|---|---|
| Most Recent Frontal/Rear Event | 4 | Front/Rear Crash | 0 | Complete (Front/Rear Page 1) |
| 1st Prior Frontal/Rear Event | 3 | Front/Rear Crash | -5120 or greater | Complete (Front/Rear Page 2) |
| Prior Frontal/Rear Event | 1 | Front/Rear Crash | N/A | Complete (Front/Rear Page 0) |

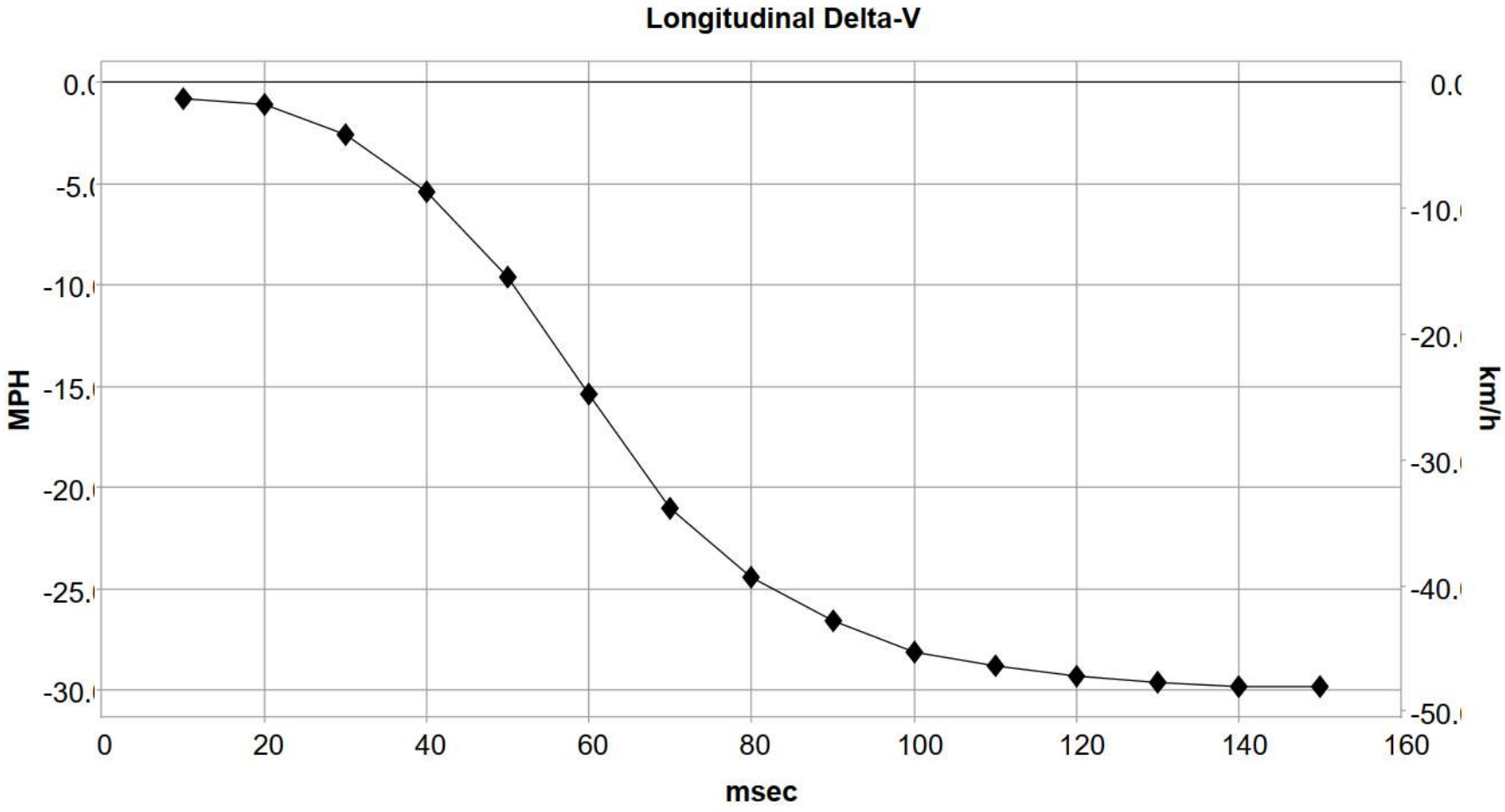
- Saves **third** front/rear event that came closest to deploying airbags (most negative DV).
- 1st prior event #3 = -5120ms ***or greater*** before most recent event, means ***NOT RELATED***
- TRG 2 was less important than TRG 1 and was overwritten when TRG 4 came in.

# Gen 1 (and 2) Max Long. $\Delta v$

- "Max" on top is the biggest number in the data table. Due to shorter duration recording, this is usually also very near the end of the crash.

**Longitudinal Crash Pulse (Most Recent Frontal/Rear Event, TRG 2 - table 1 of 2)**

| | |
|---|---|
| Max Longitudinal Delta-V (MPH [km/h]) | -29.9 [-48.1] |



# Finding PDOF in Toyota Gen1&2

- Toyota Gen 1&2 record frontal data when frontal algorithm wakes up, side data **only** when side algorithm wakes up and side data is supported.
- Many early Toyotas do **not** support side recording, you must estimate PDOF from visible crush damage.
- In later Toyotas with side recording, look for “pairs” of front and side recordings. Side recording duration is about 70ms, get PDOF by taking Y value at 70ms and X value from 70ms. BE CAREFUL – Side algorithm may wake up at a different time than frontal, **try to synchronize them.**

# Unique to Toyota

- Toyota records ADDITIONAL precrash data point at TRG and the **time** from *last regular interval* recording to TRG (resolution to 0.1 sec, 0.0 to 0.9, truncated down)
- Last data may be WRITTEN at TRG, but could still be "old" as speed data in **Gen 1 & 2** was only transmitted every 0.5 sec. If speed data did not change from prior data point, it may be old by 0.5 sec. If the last reported value is different, it cannot be older than the time from last value to prior value.

# Gen 1 & 2 Pre-Crash Data

- Accel Pedal Data in ***Voltage from sensor***
- Zero accel pedal is typically 0.78V, 100% is 3.79 V max seen to date but data limitations indicates the low and high could be different.
- You can calculate your own percentage if you want. %=(V-0.78)/(3.79-0.78) Update .5 sec
- RPM in 400 RPM increments, 400-799=400, 800-1199=800, etc. Updates every 24ms.
- Brake on/off updates immediately, no delay (accel pedal may as on when brake is on, but they are from different times)

# Unique Toyota Memory Structure

- Toyota records pre-crash in a different memory section than Delta V. It gives records "page numbers" 0 and 1 and tells you which Precrash page goes with which Delta V page.
- If there are two or more events within 0.5 sec, all may be linked to the same precrash page. This is noted in the event record summary. CDR will print the SAME precrash page 4 times
- Precrash Page 0 is stored at TRG4, not TRG7

**Event Record Summary at Retrieval**

| Events Recorded | TRG Count | Crash Type | Time (msec) | Pre-Crash & DTC Data Recording Status | Event & Crash Pulse Data Recording Status |
|---|---|---|---|---|---|
| Most Recent Event | 7 | Side Crash | 0 | Complete (Page 0) | Complete (Side Page 0) |
| 1st Prior Event | 6 | Front/Rear Crash | -20 | Complete (Page 0) | Complete (Front/Rear Page 1) |
| 2nd Prior Event | 5 | Side Crash | -228 | Complete (Page 0) | Complete (Side Page 1) |
| 3rd Prior Event | 4 | Front/Rear Crash | -282 | Complete (Page 0) | Complete (Front/Rear Page 0) |

# Un-linked precrash data

- The remaining precrash space is then known as “UNLINKED PRE CRASH” and was likely linked to a previous event. It is printed after the last linked precrash page
- In multiple event scenarios, Unlinked precrash data overlaps linked precrash and shows driver behavior farther back in time. Always check for overlap!

**System Status at Time of Retrieval**

| | |
|---|---|
| ECU Part Number | 89170-52D80 |
| ECU Generation | 06EDR |
| Recording Status, All Pages | Complete |
| Freeze Signal | ON |
| Freeze Signal Factor | Front Airbag Deployment<br>Front Pretensioner Deployment |
| Contains Unlinked Pre-Crash Data | Yes |

# Overlay Unlinked Data with Linked

**Pre-Crash Data, -5 to 0 seconds (1st Prior Event, TRG 8)**

| Time (sec) | -4.1 | -3.1 | -2.1 | -1.1 | -0.1 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 74.6 [120] | 70.8 [114] | 70.8 [114] | 69.6 [112] | 36 [58] | 36 [58] |
| Brake Switch | ON | OFF | OFF | OFF | OFF | OFF |
| Accelerator Rate (V) | 0.78 | 0.78 | 1.05 | 1.45 | 0.78 | 0.78 |
| Engine RPM (RPM) | 2,400 | 2,400 | 3,200 | 3,200 | 1,200 | 800 |

**Pre-Crash Data, -5 to 0 seconds (Unlinked, Page 1)**

| Time (sec) | -4.7 | -3.7 | -2.7 | -1.7 | -0.7 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 75.8 [122] | 75.8 [122] | 74.6 [120] | 70.8 [114] | 70.8 [114] | 69.6 [112] |
| Brake Switch | ON | ON | ON | OFF | OFF | OFF |
| Accelerator Rate (V) | 0.78 | 0.78 | 0.78 | 0.78 | 1.05 | 1.56 |
| Engine RPM (RPM) | 3,600 | 2,800 | 2,400 | 2,400 | 3,200 | 3,200 |

These two precrash data sets are from the same crash sequence

# $\Delta v$ Data in Gen 2 models

- Longer Duration Longitudinal DV (200ms vs 150ms)

  20 points @10ms intervals = 200ms duration starts at algorithm wake up - enough to get most crashes.
- Lateral DV (if equipped) for B pillar, C pillar and "Floor" (ACM) locations  25 points @4ms intervals – 100 ms duration starting 20ms before wakeup.  Only use ACM (floor) location for recon.
- Roll Angle & Lateral  Accel (if equipped).

  20 points @ 125 ms intervals beginning 625 ms before wake up

  Total duration 2.4 seconds
- Graph vs time, data table,  and "Max" single value
- Same guidelines for use as Toyota Gen 1

# Other Useful Recon Data

- Front Pass Seat Belt Buckle status
- Passenger Weight Classification (Empty/Unoccupied, Child, AF05 (small adult), **AM 50 (midsize adult *or larger*).** (*Many others only have small adult or larger class*).
- Driver Seat Position (FW forward vs RW)
- Gearshift lever position (P-R-N-Drive)
- What safety devices deployed when, deploy time marked on Delta V graph.

# Toyota Gen 2 Front Crash Report

BOSCH

CDR CRASH DATA RETRIEVAL SYSTEM

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

## CDR File Information

| | |
|---|---|
| User Entered VIN | YOUR VIN HERE |
| User | NONE |
| Case Number | NONE |
| EDR Data Imaging Date | |
| Crash Date | |
| Filename | 52690 POLICE 38.ROT.CDRX |
| Saved on | |
| Collected with CDR version | Crash Data Retrieval Tool 4.0 |
| Reported with CDR version | Crash Data Retrieval Tool 4.0 |
| EDR Device Type | airbag control module |
| Event(s) recovered | Front/Rear (2) |

Events: Front/Rear vs Side vs Rollover - & how many

## Comments

Data imported from: 52690 POLICE 38.ROT

## Data Limitations

**CDR Record Information:**

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items re
- Airbag ECU data should b
circumstances.
- If the airbags did not deplo
Non-Deployment Event. D
threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event.

called a
ecording

P. 1 Same Look/Feel as others

# System Status at Retrieval

## System Status at Time of Retrieval

| | |
|---|---|
| ECU Part Number | 89170-52690 |
| ECU Generation | 04EDR |
| Recording Status, All Pages | Complete |
| Freeze Signal | ON |
| Freeze Signal Factor | Front Airbag Deployment<br>Front Pretenstioner Deployment |
| Diagnostic Codes Exist | No |
| Total number of Events | 2 |
| Time from Previous Pre Crash TRG (msec) | 16381 or greater |
| Latest Pre-Crash Page | 0 |
| Contains Unlinked Pre-Crash Data | No |

Means record is locked, no other events can be recorded

Why record is locked

## Event Record Summary at Retrieval

| Events Recorded | TRG Count | Crash Type | Time (msec) | Pre-Crash & DTC Data Recording Status | Event & Crash Pulse Data Recording Status |
|---|---|---|---|---|---|
| Most Recent | 2 | Front/Rear Crash | 0 | Complete (Page 0) | Complete (Front/Rear Page 1) |
| 1st Prior | 1 | Front/Rear Crash | -200 | Complete (Page 0) | Complete (Front/Rear Page 0) |

Same Precrash Page

Back to back events

# System Status at Event

**System Status at Event (Most Recent, TRG 2)**

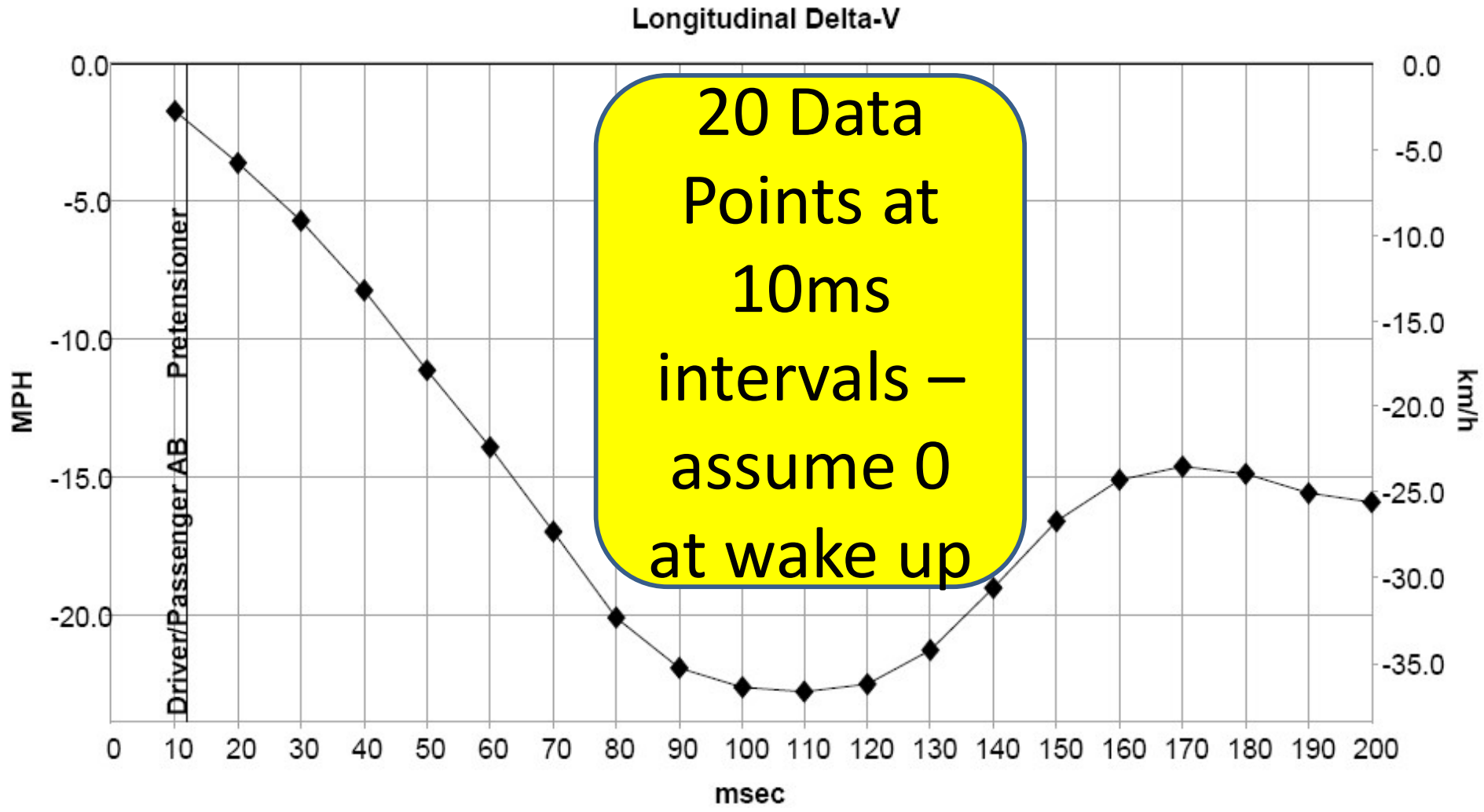
| | |
|---|---|
| Recording Status, Front/Rear Crash Info. | Complete |
| Event Type | Front/Rear Crash |
| TRG Count (times) | 2 |
| Previous Crash Type | Frontal/Rear |
| Time from Pre-Crash TRG (msec) | 200 |
| Linked Pre-Crash Page | 0 |
| Time to Deployment Command, Front Airbag, Driver (msec) | 12 |
| Time to Deployment Command, Front Airbag, Passenger (msec) | 12 |
| Event Severity Status, Driver | Level 3 |
| Event Severity Status, Passenger | N/A |
| Time to Deployment Command, Pretensioner (msec) | 12 |

Timing between airbag stages 3=short 1=long

# Delta V Data Graph

**Longitudinal Crash Pulse (Most Recent, TRG 2 - table 1 of 2)**

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Max Longitudinal Delta-V (MPH[km/h]) | -22.8 [-36.7] |



# Delta V Data Table

**Longitudinal Crash Pulse (Most Recent, TRG 2**

| Time (msec) | Longitudinal Delta-V (MPH [km/h]) |
|---|---|
| 10 | -1.7 [-2.7] |
| 20 | -3.6 [-5.8] |
| 30 | -5.7 [-9.2] |
| 40 | -8.2 [-13.2] |
| 50 | -11.1 [-17.9] |
| 60 | -13.9 [-22.4] |
| 70 | -17.0 [-27.4] |
| 80 | -20.1 [-32.3] |
| 90 | -21.9 [-35.2] |
| 100 | -22.6 [-36.4] |
| 110 | -22.8 [-36.7] |
| 120 | -22.5 [-36.2] |
| 130 | -21.3 [-34.3] |
| 140 | -19.0 [-30.6] |
| 150 | -16.6 [-26.7] |
| 160 | -15.1 [-24.3] |
| 170 | -14.6 [-23.5] |
| 180 | -14.9 [-24.0] |
| 190 | -15.6 [-25.1] |
| 200 | -15.9 [-25.6] |

20 Data Points at 10ms intervals – start at wake-up ( assume 0 at 0ms)

# Pre Crash Data - Overview

Same format Gen 1 and Gen 2

## DTCs Present at Time of Event (Most Recent, TRG 2)

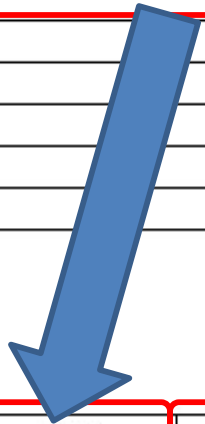
| | |
|---|---|
| Recording Status, Diagnostic | Complete |
| Ignition Cycle Since DTC was Set (times) | 0 |
| Airbag Warning Lamp ON Time Since DTC was Set (min) | 0 |
| Diagnostic Trouble Codes | None |

Some have default max value 2047

Some have default max value 15

## Pre-Crash Data, 1 Sample (Most Recent, TRG 2)

| | |
|---|---|
| Recording Status, Pre-Crash/Occupant | Complete |
| Time from Pre-Crash to TRG (msec) | 200 |
| Buckle Switch, Driver | Unbuckled |
| Buckle Switch, Passenger | Unbuckled |
| Occupancy Status, Passenger | Not Occupied |
| Seat Position, Driver | Rearward |
| Shift Position | Drive |



## Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

| Time (sec) | -4.2 | -3.2 | -2.2 | -1.2 | -0.2 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 12.4 [20] | 8.7 [14] | 5 [8] | 2.5 [4] | 1.2 [2] | 1.2 [2] |
| Brake Switch | ON | ON | ON | ON | ON | ON |
| Accelerator Rate (V) | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 |
| Engine RPM (RPM) | 800 | 800 | 800 | 400 | 800 | 800 |

# Pre Crash Data Timing

Other Points a regular 1 sec intervals

6th point Sampled at TRG

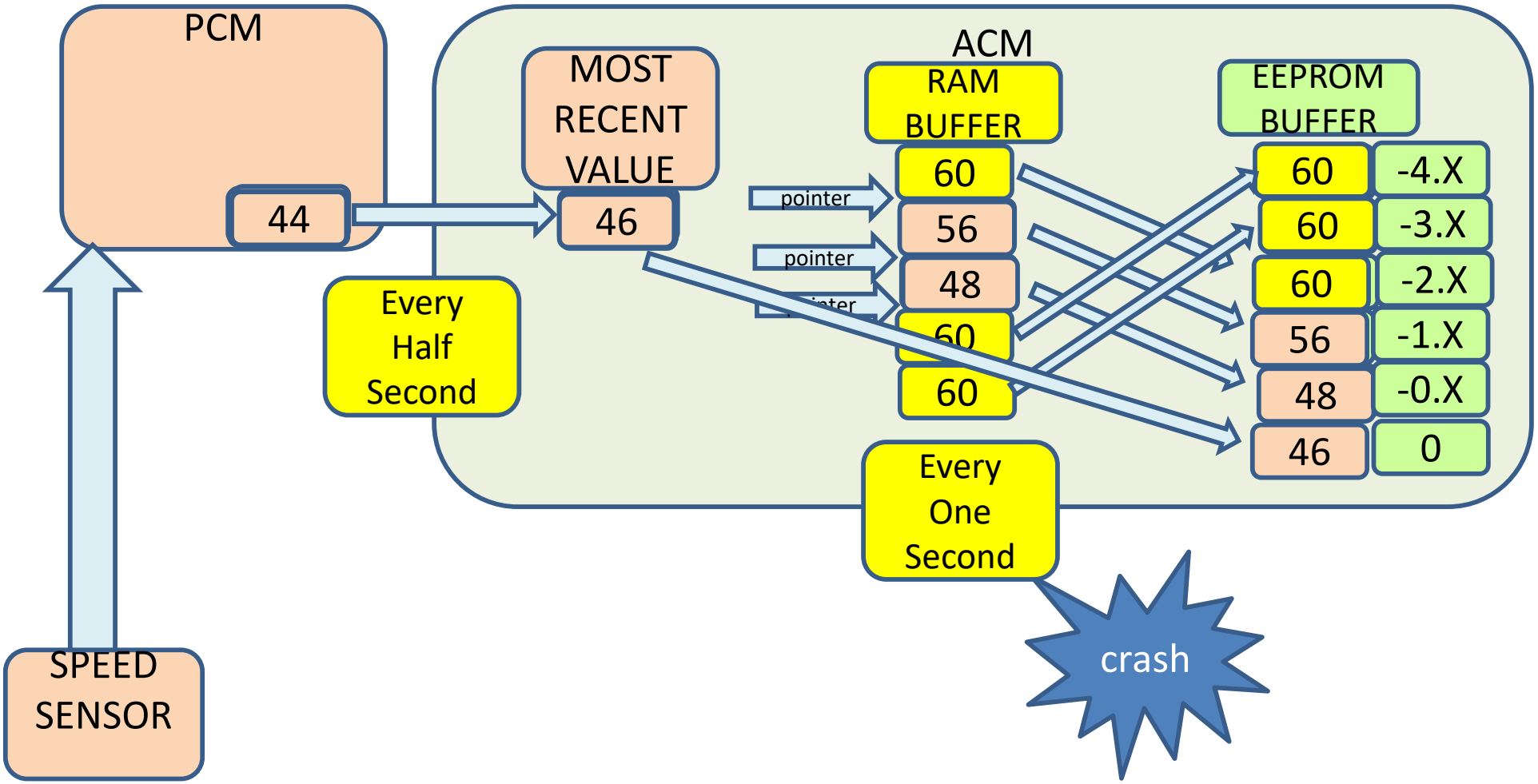
## Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

| Time (sec) | -4.2 | -3.2 | -2.2 | -1.2 | -0.2 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 12.4 [20] | 8.7 [14] | 5 [8] | 2.5 [4] | 1.2 [2] | 1.2 [2] |
| Brake Switch | ON | ON | ON | ON | ON | ON |
| Accelerator Rate (V) | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 |
| Engine RPM (RPM) | 800 | 800 | 800 | 400 | 800 | 800 |

Caution: Sampling at TRG could have un-refreshed data in buffer

New Pre Crash data will only be recorded if 500ms have passed since the last event.

# Toyota speed data transfer[34]



# Gen 1 & 2 Pre Crash Data Timing

Other Points a 1 sec intervals

6th point Sampled at TRG

## Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

| Time (sec) | -4.2 | -3.2 | -2.2 | -1.2 | -0.2 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 12.4 [20] | 8.7 [14] | 5 [8] | 2.5 [4] | 1.2 [2] | 1.2 [2] |
| Brake Switch | ON | ON | ON | ON | ON | ON |
| Accelerator Rate (V) | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 |
| Engine RPM (RPM) | 800 | 800 | 800 | 400 | 800 | 800 |

Caution: Sampling at TRG could have un-refreshed data in buffer

New Pre Crash data will only be recorded if 500ms have passed since the last event.

# Gen 1 & 2 Pre Crash Speed Data

**Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)**

| Time (sec) | -4.2 | -3.2 | -2.2 | -1.2 | -0.2 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 12.4 [20] | 8.7 [14] | 5 [8] | 2.5 [4] | 1.2 [2] | 1.2 [2] |
| Brake Switch | ON | ON | ON | ON | ON | ON |
| Accelerator Rate (V) | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 |
| Engine RPM (RPM) | 800 | 800 | 800 | 400 | 800 | 800 |

- Since the speed data buffer is only refreshed every 0.5 sec, the 0 and -0.2 data points may be from the same buffer value. If the time between TRG and the prior point is less than 0.5 sec and the values are identical, especially during precrash braking, do not assume the vehicle has stopped slowing down.
- Resolution 2km/hr (1.2 mph)
- Max Value 75.8 mph (122kph) on Gen 2 (04/06), (78.3mph/126kph in Gen 1 00/02 models)

# Gen 1 & 2 Pre Crash Accel Pedal Data

**Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)**

| Time (sec) | -4.2 | -3.2 | -2.2 | -1.2 | -0.2 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 12.4 [20] | 8.7 [14] | 5 [8] | 2.5 [4] | 1.2 [2] | 1.2 [2] |
| Brake Switch | ON | ON | ON | ON | ON | ON |
| Accelerator Rate (V) | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 |
| Engine RPM (RPM) | 800 | 800 | 800 | 400 | 800 | 800 |

- Only refreshed every 512 ms – held in buffer
- Volts not pedal percent
- Convert to % = (V-0.78)/(3.79-0.78)
- Data Limitations warns min & max may vary

# Gen 1 & 2 Pre Crash RPM Data

**Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)**

| Time (sec) | -4.2 | -3.2 | -2.2 | -1.2 | -0.2 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 12.4 [20] | 8.7 [14] | 5 [8] | 2.5 [4] | 1.2 [2] | 1.2 [2] |
| Brake Switch | ON | ON | ON | ON | ON | ON |
| Accelerator Rate (V) | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 | 0.78 |
| Engine RPM (RPM) | 800 | 800 | 800 | 400 | 800 | 800 |

- Resolution to next lower 400 rpm. Max value 5200 rpm
- If speed is reported at max 75 mph, can be used to calculate a higher min speed based on gear ratios.
- Speed in mph = RPM/trans ratio/final drive ratio/tire revs /mile*60

- Next Event

## System Status at Event (1st Prior, TRG 1)

| | |
|---|---|
| Recording Status, Front/Rear Crash Info. | Complete |
| Event Type | Front/Rear Crash |
| TRG Count (times) | 1 |
| Previous Crash Type | No Event |
| Time from Pre-Crash TRG (msec) | 0 |
| Linked Pre-Crash Page | 0 |
| Time to Deployment Command, Front Airbag, Driver (msec) | Not Commanded |
| Time to Deployment Command, Front Airbag, Passenger (msec) | Not Commanded |
| Event Severity Status, Driver | N/A |
| Event Severity Status, Passenger | N/A |
| Time to Deployment Command, Pretensioner (msec) | Not Commanded |

# RTFDL Caution

- In the 00 EDR, seat belt status in ND events is the status at time of READOUT, not the time of EVENT.
- This is duly noted in the data limitations, but has tripped some people up. A module was read in car that said unbelted, then re-read on the benchtop that gave the default value of belted.

- For Non-Deployment Event, "Buckle Switch", "Ignition Cycle Since DTC was Set" and "Airbag Warning Lamp ON Time Since DTC was Set" is the ACM parameter data at the time of retrieval, not the parameter data when the recording trigger was established. For Deployment Event, it is the information when the airbag deployment.

# Bug in Gen 1&2 Long. Delta V

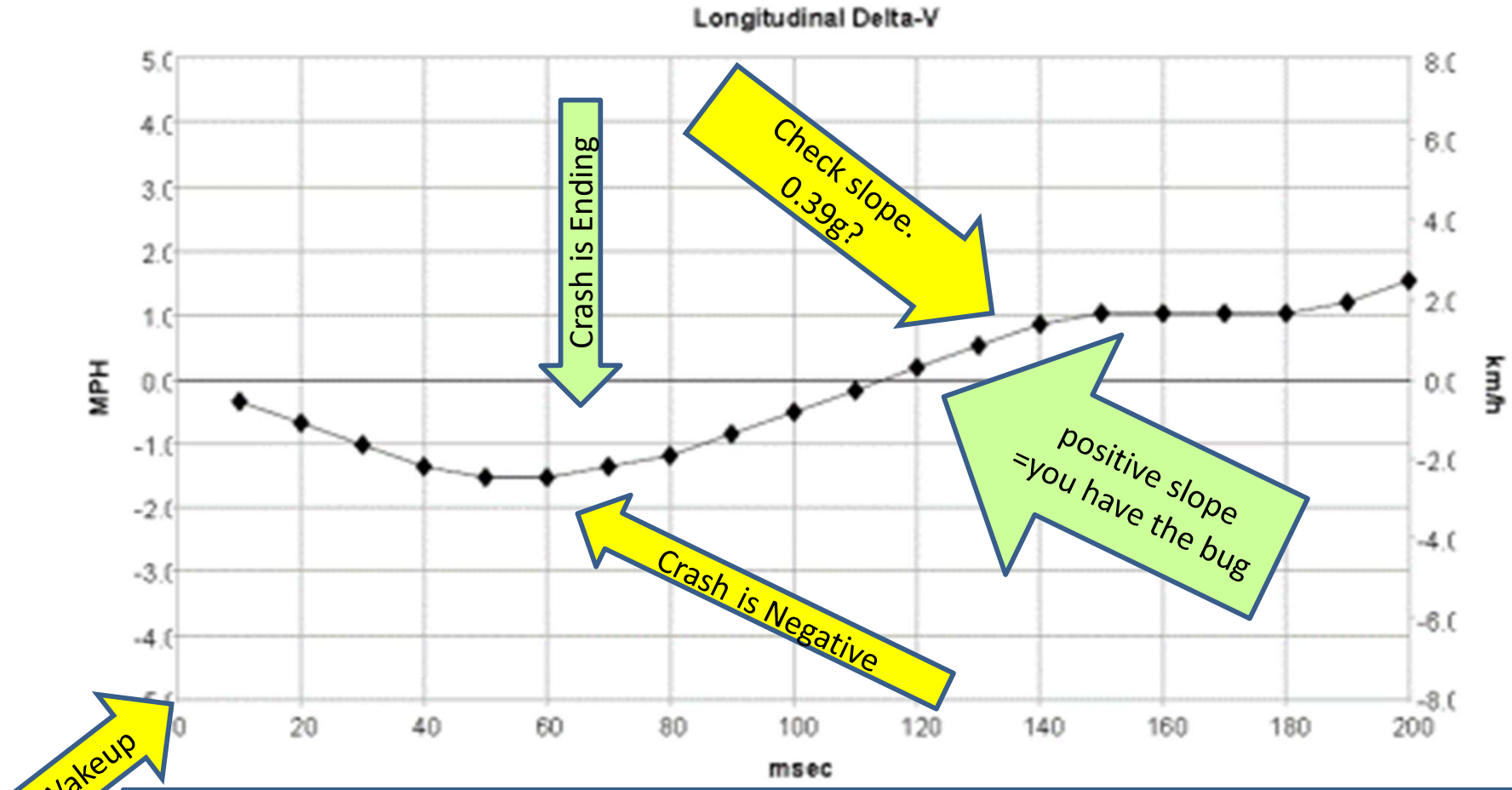
- Empirical research indicates in ***all Gen 1's*** tested to date and ***SOME Gen 2*** models
- ***Under*** reports *frontals* slightly
- ***Over*** reports *rears* slightly
- Sources in Toyota indicate the zero-offset applied when converting analog to digital accelerometer output may be off by 1 count, creating an error of 0.39G.
- 0.39g*150msec*32.2/1.466 = 1.3 mph
- Empirical data suggests 2 counts on some
- SAE papers from MEA discussing this include

  2017-01-1438, 2016-01-1494, 2013-01-1268

# How to Identify you have the bug

**Longitudinal Crash Pulse (Most Recent Event, TRG 3 - table 1 of 2)**

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Max Longitudinal Delta-V (MPH [km/h]) | 1.5 [2.5] |




Adjust Max value by 0.39g * time at max (or amount based on slope)

No positive slope = no adjustment needed

# Toyota Gen 2 Side Crash Report

BOSCH

CDR CRASH DATA RETRIEVAL SYSTEM

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

## CDR File Information

| | |
|---|---|
| User Entered VIN | 4T1BF3EK4BU664098 |
| User | Richard R. Ruth, P.E. |
| Case Number | Gazdick salvage yard find |
| EDR Data Imaging Date | 06/01/2011 |
| Crash Date | |
| Filename | 4T1BF3EK4BU664098_ACM_GAZDICK_BODY_SHOP.CDRX |
| Saved on | Wednesday, June 1 2011 at 21:52:48 |
| Collected with CDR version | Crash Data Retrieval Tool 4.0 |
| Reported with CDR version | Crash Data Retrieval Tool 4.0 |
| EDR Device Type | airbag control module |
| Event(s) recovered | Front/Rear (1), Side (1) |

## Comments

imaged direct to module with cable 614 and 387 adaptor

## Data Limitations

**CDR Record Information:**

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If the airbags did not deploy or the pretensioners did not operate during an event that meets a specified recording threshold, it is called a Non-Deployment Event. Data from a Non-Deployment Event can be overwritten by a succeeding event that meets the specified recording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event

Events: Front/Rear vs Side vs Rollover - & how many

# Side Event Polarity

## Which Side were you hit on?

**System Status at Event (Most Recent, TRG 2)**

| | |
|---|---|
| Recording Status, Side Crash Info. | Complete |
| Event Type | Side Crash |
| TRG Count (times) | 2 |
| Recorded Side | Passenger's Side |
| Previous Crash type | Frontal/Rear |
| Time from Pre-Crash TRG (msec) | 24 |
| Linked Pre-Crash Page | 0 |
| Time to Deployment Command, Row 1 (msec) | Not Commanded |
| Time to Deployment Command, Row 2 (msec) | Not Commanded |

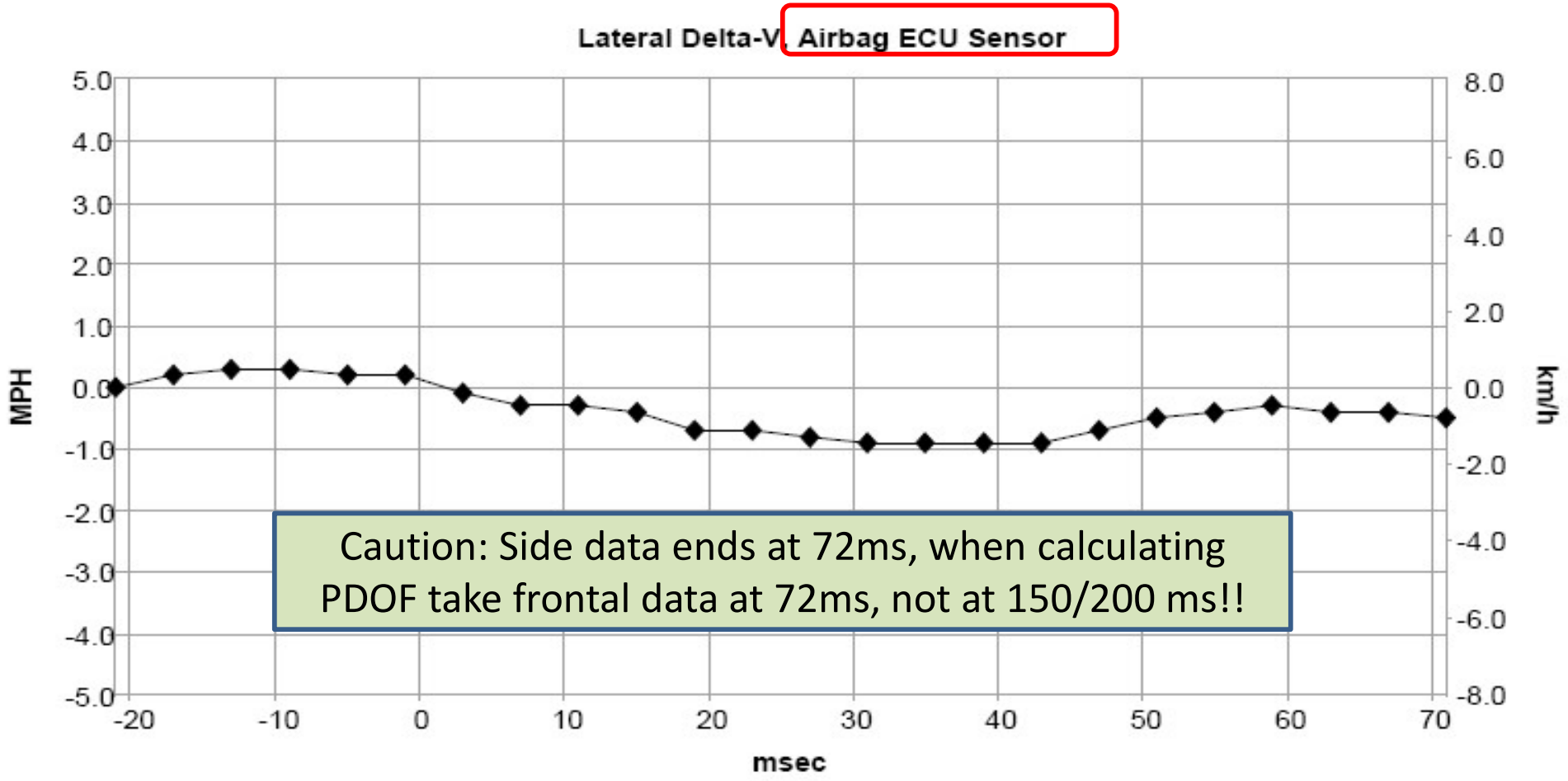
- Side contacted is NOT determined by sign.
- Recorded Side is displayed.

# Gen 1&2: Lateral Delta V Graph

Floor Sensor means ACM (same as Airbag ECU) – closest to CG

**Lateral Crash Pulse (Most Recent, TRG 2 - table 1 of 2)**

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Time from TRG to Next Sample (msec) | 3 |
| Max Lateral Delta-V, Floor (MPH[km/h]) | -0.9 [-1.4] |
| Max Lateral Delta-V, B-Pillar Sensor (MPH[km/h]) | 1.5 [2.4] |
| Max Lateral Delta-V, C-Pillar Sensor (MPH[km/h]) | 2.7 [4.3] |



# Gen 1&2: Use ECU Side Data Only

**Lateral Crash Pulse (Most Recent, TRG 2 - table 2 of 2)**

| Time (msec) | Lateral Delta-V, Airbag ECU Sensor (MPH [km/h]) | Lateral Delta-V, B-Pillar Sensor MPH [km | Lateral Delta-V, C-Pillar Sensor H [km/h] |
|---|---|---|---|
| -21 | 0.0 [0.0] | 0.0 [0. | 0 [0.0] |
| -17 | 0.2 [0.3] | 0.1 [0 | [0.3] |
| -13 | 0.3 [0.5] | 1 [ | 0.5 |
| -9 | 0.3 [0.5] | 1 | 0 |
| -5 | 0.2 [0.3] | | |
| -1 | 0.2 [0.3] | | |
| 3 | -0.1 [-0.2] | | |
| 7 | -0.3 [-0.5] | | |
| 11 | -0.3 [-0.5] | | 1 |
| 15 | -0.4 [-0.6] | | |
| 19 | -0.7 [-1.1] | | |
| 23 | -0.7 [-1.1] | | |
| 27 | -0.8 [-1.3] | | |
| 31 | -0.9 [-1.4] | | |
| 35 | -0.9 [-1.4] | 3 | 3. |
| 39 | -0.9 [-1.4] | .3 [2 | [3.4] |
| 43 | -0.9 [-1.4] | 1.3 [2 | [3.9] |
| 47 | -0.7 [-1.1] | 1.5 [2.4 | 5 [4.0] |
| 51 | -0.5 [-0.8] | 1.5 [2.4] | .6 [4.2] |
| 55 | -0.4 [-0.6] | 1.5 [2.4] | 2.7 [4.3] |
| 59 | -0.3 [-0.5] | 1.5 [2.4] | 2.7 [4.3] |
| 63 | -0.4 [-0.6] | 1.4 [2.3] | 2.7 [4.3] |
| 67 | -0.4 [-0.6] | 1.4 [2.3] | 2.7 [4.3] |
| 71 | -0.5 [-0.8] | 1.5 [2.4] | 2.7 [4.3] |

Pre Crash Data Page Gen 1& 2 (looks **same** for side and frontal)
Check page numbers, if page # is same they are identical

## DTCs Present at Time of Event (Most Recent, TRG 2)

| | |
|---|---|
| Recording Status, Diagnostic | Complete |
| Ignition Cycle Since DTC was Set (times) | 0 |
| Airbag Warning Lamp ON Time Since DTC was Set (min) | 0 |
| Diagnostic Trouble Codes | None |

## Pre-Crash Data, 1 Sample (Most Recent, TRG 2)

| | |
|---|---|
| Recording Status, Pre-Crash/Occupant | Complete |
| Time from Pre-Crash to TRG (msec) | 700 |
| Buckle Switch, Driver | Buckled |
| Buckle Switch, Passenger | Unbuckled |
| Occupancy Status, Passenger | Not Occupied |
| Seat Position, Driver | Rearward |
| Shift Position | Drive |

## Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

| Time (sec) | -4.7 | -3.7 | -2.7 | -1.7 | -0.7 | 0 (TRG) |
|---|---|---|---|---|---|---|
| Vehicle Speed (MPH [km/h]) | 46 [74] | 47.2 [76] | 47.2 [76] | 47.2 [76] | 42.3 [68] | 24.9 [40] |
| Brake Switch | OFF | OFF | OFF | OFF | ON | ON |
| Accelerator Rate (V) | 1.25 | 1.09 | 0.98 | 0.98 | 0.78 | 0.78 |
| Engine RPM (RPM) | 1,600 | 1,200 | 1,200 | 1,200 | 800 | 800 |

# Speed at Impact Worksheet GEN 1 & 2

TWO changes from 42 to 24?

| PRE-CRASH DATA | | | | | | |
|---|---|---|---|---|---|---|
| Time | -4.7 | -3.7 | -2.7 | -1.7 | -0.7 | 0 |
| Speed | 46 (74) | 47.2 (76) | 47.2 (76) | 47.2 (76) | 42.3 (68) | 24.9 (40) |
| Brake | Off | Off | Off | Off | On | On |
| Time Range | -5.2 to -4.7 | -4.2 to -3.7 | -3.2 to -2.7 | -2.2 to -1.7 | -1.2 to -0.7 | 0.2 to 0.0 |

| SPEED AT IMPACT WORKSHEET | | English | | Metric | |
|---|---|---|---|---|---|
| | | MIN | MAX | MIN | MAX |
| Last Good Data Point mph (km/h) | | 24.9 | 24.9 | 40 | 40 |
| Braking | -0.2 sec* 15 mph/sec | -3.0 | 0 | -4.8 | 0 |
| Slip +5% | | 1.2 | 1.2 | 2 | 2 |
| Speedo +/-4% | | -1 | 1 | -1.6 | 1.6 |
| Range | | 22.1 | 27.1 | 35.6 | 43.6 |

# Age of TRG data sample Gen 1 & 2

| Time Last Point to TRG | Changed twice | Changed once | Didn't change |
|---|---|---|---|
| 0 | n/a | 0 | 0.5 |
| 0.1 | n/a | 0.1 | 0.5 |
| 0.2 | n/a | 0.2 | 0.5 |
| 0.3 | n/a | 0.3 | 0.5 |
| 0.4 | n/a | 0.4 | 0.5 |
| 0.5 | 0 | 0.5 | n/a |
| 0.6 | 0.1 | 0.5 | n/a |
| 0.7 | 0.2 | 0.5 | n/a |
| 0.8 | 0.3 | 0.5 | n/a |
| 0.9 | 0.4 | 0.5 | n/a |

- Data at TRG can be 0.5 sec old, but may be LESS than 0.5 sec if data has changed ONCE from prior sample within 0.5 sec (9 mph or less)
- Data at TRG can ALSO be LESS than 0.5 sec if data has changed TWICE from prior sample (10 mph or more)
- Version 2.01 of the speed at impact worksheet excel program does this for you

# Gen 3 -Part 563 Intent Toyota”12EDR”

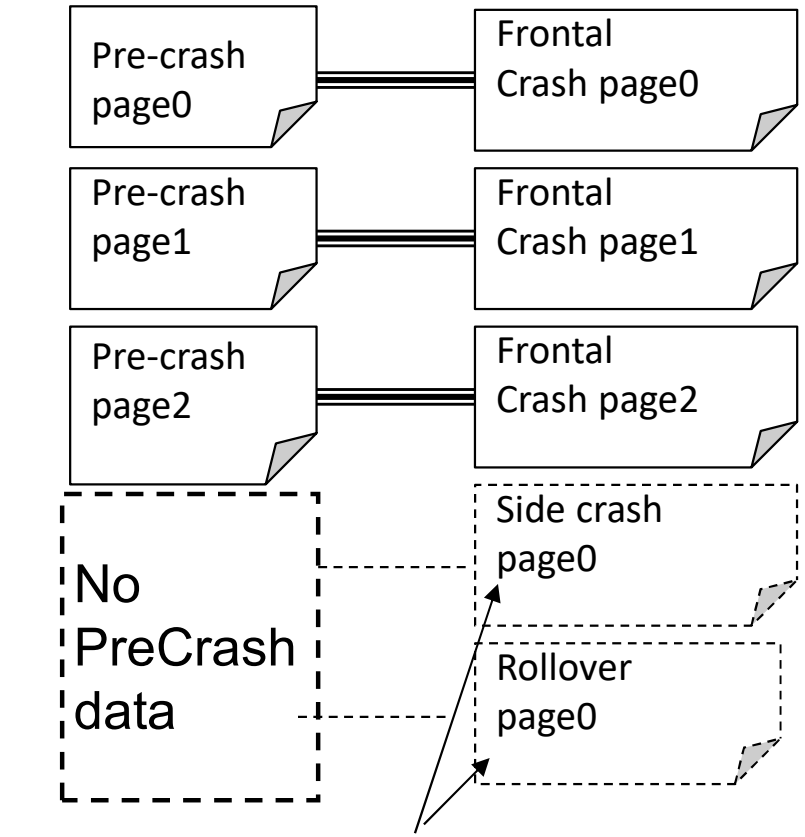
- Phase in mid 2012 MY
- Adds key cycles, deploy times, longer ΔV (250ms), more pre-crash 5 sec @ 0.5 sec intervals, speed to 124 mph (200 KPH), Accel %.
- Lat ΔV only recorded in side events (Unique to Toyota) and ONLY FROM SIDE SATELLITES, not ACM
- Recording threshold 5mph ΔV min. for frontal event
- Side recording threshold stays at 3.5g wakeup
- Adds Throttle %, Steering, Yaw Rate, Brake Pressure, VSC Long Accel, Cruise, Gear, RPM @ 100 rpm resolution.

# Gen 3 “12 EDR”

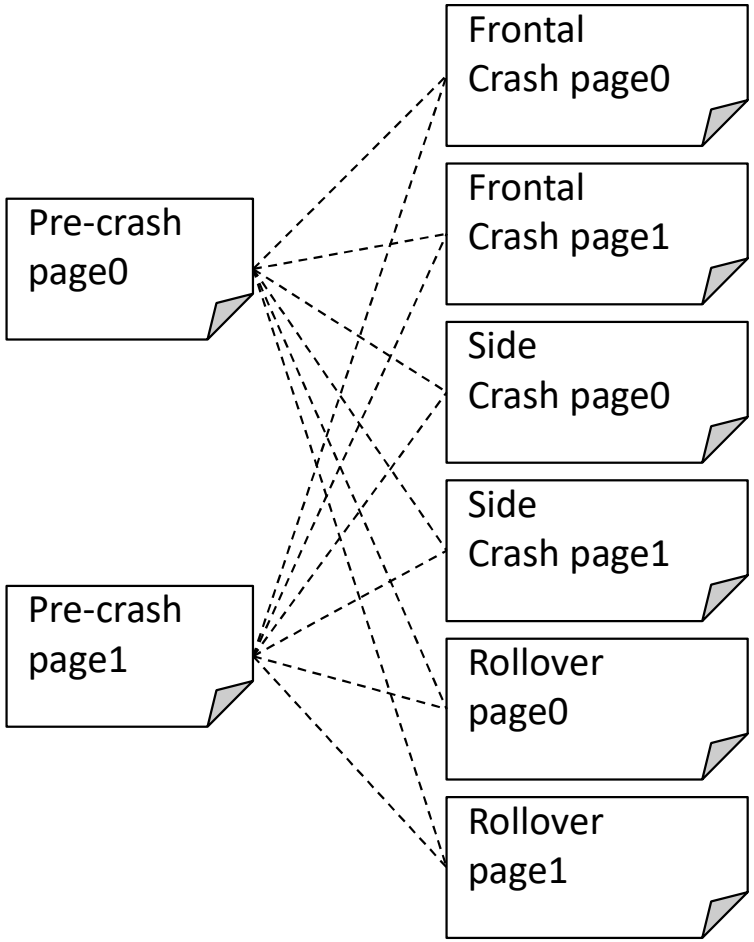
- Speed data updated more frequently now - every 20 to 24 ms – (64 ms on accel pedal)
- Toyota continues to sample data point at TRG
- Brake **pressure** data element added in MPa
- Accel pedal data in % vs. volts
- Analyze the same as any other 563 intent module, except recognize that last data point is for all practical purposes **at TRG** now.

# Toyota Memory "Pages"

| 00/02EDR (GEN 1) | 04/06EDR (GEN 2) and **12EDR (GEN 3)** |
|---|---|
| ▪ Frontal Crash: **3** pages<br>▪ Side and Rollover: 1page (no Pre-Crash) | ▪ Pre-Crash : **2** pages<br>▪ Frontal Side Rollover :2 pages each |




Not all vehicles have Side and/or Rollover




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# Pre-Crash Parameter Comparison

| Parameter | 04/06 EDR | | 12 EDR | |
|---|---|---|---|---|
| Vehicle Speed (Part 563) | 2 km/h /LSB | 0~122 [km/h] | 1 km/h /LSB | 0 ~ 200 [km/h] |
| Accelerator Pedal (Part 563) | .039 V/LSB | 0~5 [V] | 0.5 %/LSB | 0 ~ 100 [%] |
| Engine RPM (Part 563) | 400 RPM/LSB | 0~5200 [RPM] | 100 RPM/LSB | 0 ~ 12800 [RPM] |
| Service Brake (Part 563) | ON/OFF | | ON/OFF | |
| Shift Position | R / N/ P / D | | R / N / P / D / 5 / 4/ 3 / 2/ LO/ B/ Seq | |
| Percentage of Engine Throttle | | | 0.5%/LSB | 0 ~ 100 [%] |
| Motor RPM (Part 563) | | | 100 RPM/LSB | -7500 ~ 17500 [RPM] |
| Brake Oil Pressure | | | 0.048 MPa | 0 ~ 12.144 [MPa] |
| Sequential Shift | | | 1 / 2 / 3 /4 / 5 / 6 / 7 / 8 | |
| Steering Input (Part 563) | | | 3 deg/LSB | -375 ~ 375 [deg] |
| VSC Yaw Rate | | | 0.488 deg/s/LSB | -61 ~ 61 [deg/s] |
| VSC Longitudinal Acceleration | | | 0.07178 $m/s^2$/LSB | -8.9725~8.9725 [$m/s^2$] |
| Cruise Control Status | | | ON/OFF | |
| Drive Mode - Power<br>Drive Mode – Eco<br>Drive Mode - Sport<br>Drive Mode - Snow<br>Drive Mode - EV | | | ON/OFF | |

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# Gen 3 Longitudinal Stability Control Accel Data

- Units are in $m/sec^2$, divide by 9.8 to get G's (dividing by 10 gets you close)
- Data is like a Vericom, similar to Ford stability control system data, just not as frequent.
- The max value is 8.973 $m/sec^2$.
- The last value at TRG is often 8.973, it may be sampled as the crash is beginning. Do not assume max braking from last data point alone.

# Getting PDOF from 12 EDR

- 12 EDR records side Delta V ONLY if side algorithm wakes up, and ONLY from side satellites – NO data from ACM!!!!!
- In Gen 1&2, we told you to IGNORE B and C pillar data and use ACM data only.
- With NO DATA from ACM recorded, your only choice is to use B pillar data (closest sensor to CG)
- Take side Delta V at end of trace (typically 70ms) and take arc tangent of Y over X at same time (70ms).
- Try to synchronize X and Y pulses based on relative wake-up

# New “13 EDR” & “15 EDR”

- 13 EDR Adds lateral Accel data from ACM (“floor sensor”) in ***front/rear*** **events, synchronized to Long DV time 0** (Fixes biggest drawback to 12 EDR),
- Lateral acceleration data table is in meters/sec$^2$, calculate Delta V from it yourself. For each point, take **(accel/9.81)*(32.2/1.466)*time interval**, and add them to get the cumulative value in **mph**.
- Metric, take accel*3.6*time interval, add them to get the cumulative value in kph.
- See “Toyota 13 EDR” **TEMPLATE** in student materials to do this math for you calculates both Delta V and PDOF.
- Side events are still recorded separately in side wakeups, & record 70ms of side accel, but are NO LONGER NEEDED for analysis if the frontal algorithm triggered and recorded with side data. .
- 15 EDR adds side impact door pressure sensor. Pressure sensor data is NOT used in reconstruction analysis, it is just for quicker side airbag activation.

# 13 EDR Long DV & Lat Accel

**Longitudinal/Lateral Crash Pulse (3rd Prior Event, TRG 1 - table 2 of 2)**

| Time (msec) | Longitudinal Delta-V (MPH [km/h]) | Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2) | |
|---|---|---|---|
| 0 | 0.0 [0.0] | 0.0 | |
| 10 | -1.1 [-1.7] | 4.6 | ON |
| 20 | -2.0 [-3.3] | 7.7 | ON |
| 30 | -6.6 [-10.6] | 41.4 | ON |
| 40 | -7.8 [-12.5] | 96.5 | ON |
| 50 | -8.6 [-13.8] | 19.9 | ON |
| 60 | -9.1 [-14.7] | -79.7 | ON |
| 70 | -9.8 [-15.8] | 18.4 | ON |
| 80 | -11.3 [-18.1] | -16.9 | ON |
| 90 | -12.8 [-20.6] | 64.4 | ON |
| 100 | -13.8 [-22.2] | 23.0 | ON |
| 110 | -15.5 [-25.0] | 30.6 | ON |
| 120 | -17.6 [-28.3] | -49.0 | ON |
| 130 | -19.8 [-31.8] | 13.8 | ON |
| 140 | -22.7 [-36.5] | 24.5 | ON |
| 150 | -22.3 [-35.9] | -76.6 | ON |
| 160 | -23.3 [-37.5] | 52.1 | ON |
| 170 | -24.0 [-38.6] | 6.1 | ON |
| 180 | -24.7 [-39.8] | -13.8 | ON |
| 190 | -25.6 [-41.2] | -12.3 | ON |
| 200 | -26.1 [-42.1] | 9.2 | ON |
| 210 | -27.0 [-43.5] | 29.1 | ON |
| 220 | -27.5 [-44.2] | 26.0 | ON |
| 230 | -27.7 [-44.6] | 19.9 | ON |
| 240 | -27.9 [-44.9] | 7.7 | ON |
| 250 | -28.1 [-45.2] | 4.6 | ON |

Save as CSV
Integrate to get $\Delta v$

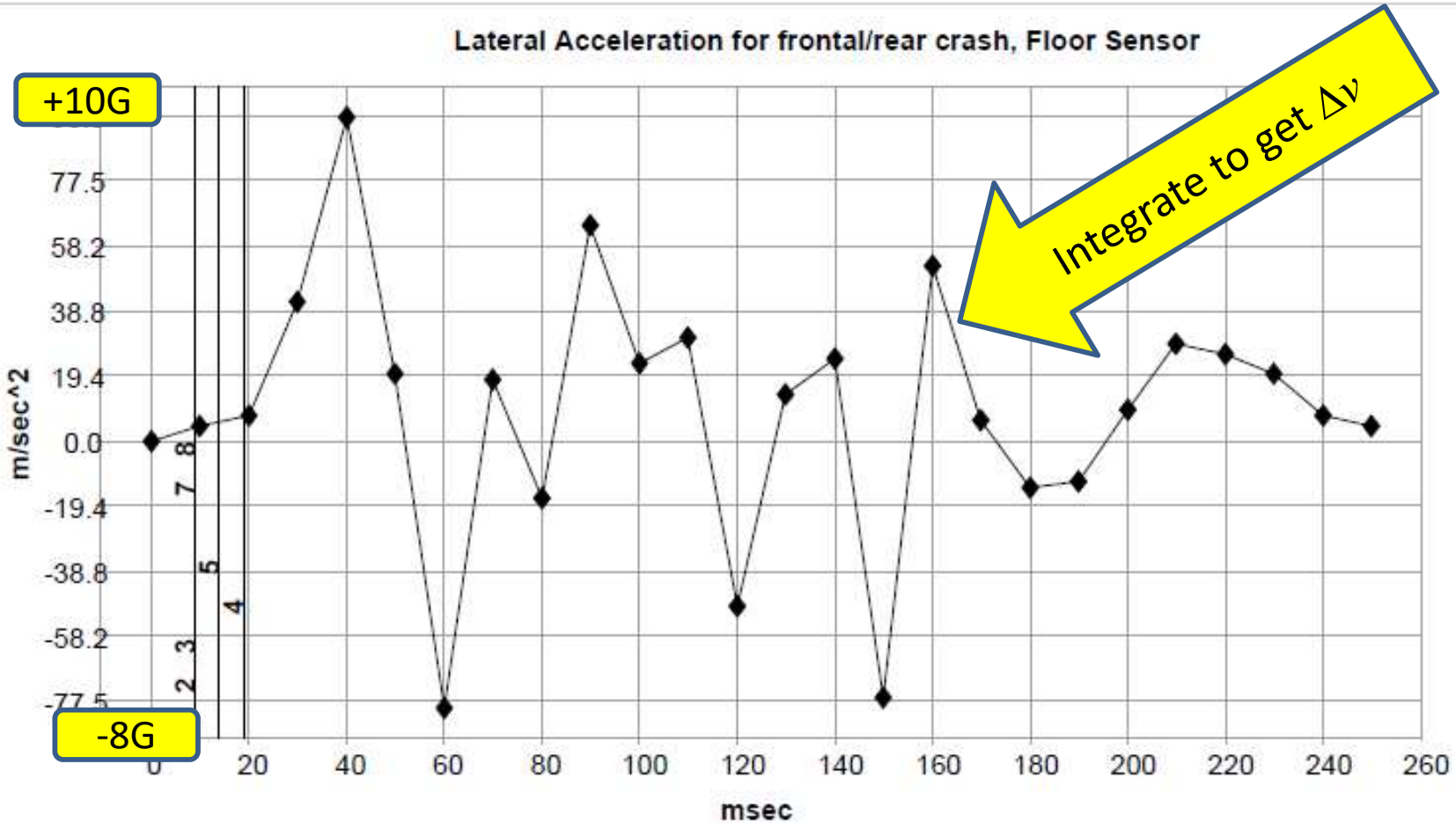
# Template in Student Ref Materials

**Toyota 13 EDR Lat Accel to Lat DV Template**

INSTRUCTIONS
While in CDR save data is CSV format
Copy Columns A thru E from CSV file for 0-250ms
Paste into Column A thru E below. D is metric and is hidden

| LONGITUDINAL/LATERAL | | PASTE DATA BELOW | | CALCULATIONS - IMPERIAL | | | CALCUALTIONS - METRIC | | |
|---|---|---|---|---|---|---|---|---|---|
| Time (msec | Longitudinal Delta-V (MPH) | Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2) | Power Supply Status | Lat DV from Floor Sensor, (MPH) | PDOF in degrees | Total DV | Long DV (KPH) | Lat DV (kph) | Total DV (kph) |
| 0 | 0 | 0 | ON | 0 | | 0.0 | 0.0 | 0.0 | 0.0 |
| 10 | -1.2 | -5.7 | ON | -0.1 | 6.1 | 1.2 | -1.9 | -0.2 | 1.9 |
| 20 | -3.6 | -11.5 | ON | -0.4 | 6.1 | 3.6 | -5.8 | -0.6 | 5.8 |
| 30 | -6.7 | -17.2 | ON | -0.8 | 6.6 | 6.7 | -10.8 | -1.2 | 10.9 |
| 40 | -9.9 | -70.9 | ON | -2.4 | 13.4 | 10.2 | -16.0 | -3.8 | 16.4 |
| 50 | -13 | -113 | ON | -4.9 | 20.6 | 13.9 | -21.0 | -7.9 | 22.3 |
| 60 | -15.6 | -51.7 | ON | -6.1 | 21.2 | 16.7 | -25.1 | -9.7 | 26.9 |
| 70 | -16.5 | -34.5 | ON | -6.8 | 22.5 | 17.9 | -26.5 | -11.0 | 28.7 |
| 80 | -17.3 | -7.7 | ON | -7.0 | 22.0 | 18.7 | -27.9 | -11.3 | 30.0 |
| 90 | -17.5 | -15.3 | ON | -7.3 | 22.8 | 19.0 | -28.1 | -11.8 | 30.5 |
| 100 | -17.7 | -17.2 | ON | -7.7 | 23.6 | 19.3 | -28.4 | -12.4 | 31.1 |
| 110 | -18 | 0 | ON | -7.7 | 23.2 | 19.6 | -29.0 | -12.4 | 31.5 |
| 120 | -18.2 | 1.9 | ON | -7.7 | 22.9 | 19.8 | -29.2 | -12.4 | 31.8 |
| 130 | -18.3 | 0 | ON | -7.7 | 22.8 | 19.8 | -29.5 | -12.4 | 31.9 |
| 140 | -18.3 | 0 | ON | -7.7 | 22.8 | 19.8 | -29.5 | -12.4 | 31.9 |
| 150 | -18.3 | 3.8 | ON | -7.6 | 22.5 | 19.8 | -29.5 | -12.2 | 31.9 |
| $grey160 | $grey0.0 | $grey0.0 | $greyON | #VALUE! | #VALUE! | **#VALUE!** | **$grey0.0** | #VALUE! | #VALUE! |

# 13 EDR lat accel graph from 28 mph (big) longitudinal event




Deployment Time Marker Key

| | |
|---|---|
| 1 | Driver Airbag Deployment Time |
| 2 | Passenger Airbag Deployment Time |
| 3 | Driver/Passenger Pretensioner |
| 4 | Driver 2nd Stage Airbag Deployment Time |
| 5 | Passenger 2nd Stage Airbag Deployment |
| 6 | Driver/Passenger AHR |
| 7 | Driver CSA |
| 8 | Passenger CSA |
| 9 | Rear Window Airbag Deployment Time |

# 13 EDR Accelerometer Range Based on an 86 mph Delta V case

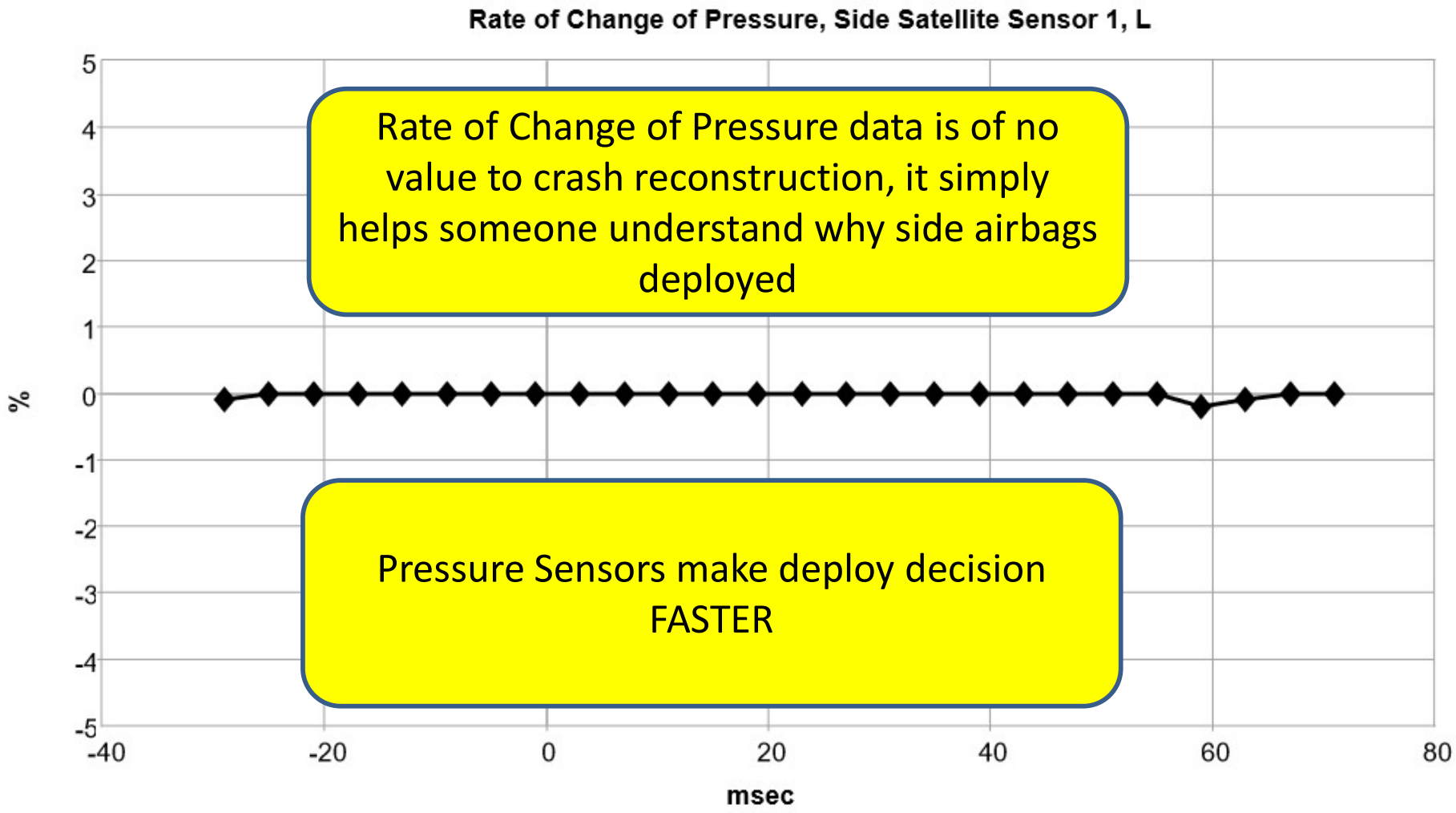
| | |
|---|---|
| Clipping Time of Longitudinal Delta-V (msec) | 59.0 |
| Clipping Time of Lateral Acceleration, Floor Sensor (msec) | 22.0 |

**Longitudinal/Lateral Crash Pulse (3rd Prior Event, T**

| Time (msec) | Longitudinal Delta-V (MPH [km/h]) | Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2) |
|---|---|---|
| 0 | 0.0 [0.0] | 0.0 |
| 10 | -2.6 [-4.1] | 33.7 |
| 20 | -7.9 [-12.7] | 27.6 |
| 30 | -17.5 [-28.1] | -99.6 |
| 40 | -25.3 [-40.6] | -39.8 |
| 50 | -34.0 [-54.7] | 39.8 |
| 60 | -51.7 [-83.1] | 137.9 |
| 70 | -67.8 [-109.0] | 165.5 |
| 80 | -73.2 [-117.9] | 125.6 |
| 90 | -75.7 [-121.8] | -46.0 |
| 100 | -77.4 [-124.5] | -18.4 |

Longitudinal – Clip at 59ms - 50ms to 60 ms incremental Delta V = 17.7 mph
(17.7MPH * 1.466/32.2)/.01 SEC = 80.5G
Lateral – Clip at 22ms, -99.6 @ 30msec = 10G, not likely
Highest Measured Value (165.5 m/sec2 @ 70ms)/9.81=16.9g
Conclusion: Longitudinal and Lateral Accels are different ranges

# Toyota 15 EDR
# side impact door PRESSURE sensors



# New “17 EDR”

- “Front/Rear” events also triggered by 5mph Y $\Delta v$ or side deployment
- Y $\Delta v$ from ACM added to frontal events
- More precrash registers (6 vs 2 prior)
- 4 front/rear & 4 side $\Delta v$ registers, 2 rollover
- Separate side events still reported, but ***do NOT have precrash associated with them***. Side events have detail of side satellite sensors (generally not used by Recons).
- Odometer data element added

# 17 EDR Event Record Summary
# more events, detail on recording status

## Event Record Summary at Retrieval

| Events Recorded | TRG Count | Crash Type | Time (msec) | Pre-Crash Recording Status | Diagnostic Data Recording Status | Occupant Data Recording Status | Crash Info Recording Status | Time Series Recording Status |
|---|---|---|---|---|---|---|---|---|
| Most Recent Event | 10 | Side Crash | 0 | N/A | Complete | N/A | | Complete |
| 1st Prior Event | 9 | Frontal/Rear/Side | -898 | Complete | | Complete | Complete | Complete |
| 2nd Prior Event | 8 | Side Crash | -925 | N/A | | | N/A | Complete |
| 3rd Prior Event | 6 | Side Crash | N/A | | | N/A | N/A | Complete |
| 4th Prior Event | 5 | Rollover | N/A | Complete | Complete | | Incomplete | Incomplete |
| 5th Prior Event | 4 | Frontal/Rear/Side Crash | N/A | Complete | | Complete | Complete | Complete |
| 6th Prior Event | 3 | Side Crash | N/A | N/A | Complete | N/A | N/A | Complete |
| 7th Prior Event | 2 | Rollover | N/A | Complete | Complete | Complete | Complete | Complete |

Pages NOT numbered like Gen3

#7 missing

Lost count due to 7 missing

No precrash for sides

# 17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash.

| | |
|---|---|
| Event(s) recovered | Front/Rear/Side Events (2), Side Events (4), Rollover Events (2) |

Note Severity of Each Event in the summary

## Event Record Summary at Retrieval

| Events Recorded | TRG Count | Crash Type | Time (msec) | Pre-Crash Recording Status | Diagnostic Data Recording Status | Occupant Data Recording Status | Crash Info Recording Status | Time Series Recording Status |
|---|---|---|---|---|---|---|---|---|
| Most Recent Event | 13 | Side Crash | 0 | N/A | Complete | N/A | N/A | Complete |
| 1st Prior Event | 12 | 146° Rollover | N/A | Complete | Complete | Complete | Complete | Complete |
| 2nd Prior Event | 10 | Side Crash | N/A | N/A | Complete | N/A | N/A | Complete |
| 3rd Prior Event | 9 | Frontal/Rear/Side Crash | N/A | Complete | Complete | Complete | Complete | Complete |
| 4th Prior Event | 8 | Side Crash | N/A | N/A | Complete | N/A | N/A | Complete |
| 5th Prior Event | 7 | 200° Rollover | Deployment | Complete | Complete | Complete | Complete | Complete |
| 6th Prior Event | 5 | Side Crash | N/A | N/A | Complete | N/A | N/A | Complete |
| 7th Prior Event | 3 | Frontal/Rear/Side Crash | -38 DVx, -20 DVy<br>N/A<br>Deployment | Complete | Complete | Complete | Complete | Complete |

3 key cycles after 12

-3 DVx, -6DVy

#11 skipped?

Lost at #13

#6 side overwritten?

#4 side overwritten?

#3 Says #2 was not related (so we didn't miss anything important)

# 17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash. Compare Time Count at Event to find relative time of events

Deployment

Deployment

#3 Big hit -38 DVx

## System Status at Event (7th Prior Event, TRG 3)

| | | | | | | |
|---|---|---|---|---|---|---|
| TRG Count (times) | 3 | 5 | 7 | 8 | 9 |
| Event Type | Frontal/Rear/Side Crash | Side Crash | Rollover | Side Crash | Frontal/Rear/Side Crash |
| Previous Crash Type | No Event | Side Crash | Side Crash | Rollover | Side Crash |
| Time from Previous TRG (msec) | 32767 or greater | 104.5 | 32.5 | 916.0 | 50.0 |
| Time from Time Zero to TRG (msec) | 1.5 | ON | ON | OFF | 46.5 |
| Event Establishment Factor | Frontal Crash | loyment, Driver / Front Airbag Deployment, Passenger | llover CSA Deployment | None | Side Crash, Right |
| TRG Establishment Factor | Frontal Crash | Complete | Complete | Complete | Side Crash, Right |
| Freeze Signal | ON | 45,240 [72,807] | 45,240 [72,807] | 45,240 [72,807] | OFF |
| Freeze Signal Factor | | 4431 | 4431 | 4431 | None |
| Recording Status, Front/Rear and Side Crash Info. | Complete | 1,711,600 | 1,713,500 | 1,714,300 | 45,240 [72,807] |
| Odometer signal (miles [km]) | 45,240 [72,807] | Normal | Normal | Normal | 4431 |
| Trip count (times) | 4431 | | | | 1,714,400 |
| Time count (msec) | 1,711,500 | | | | Normal |
| Time count input system | Normal | | | | |

## Deployment Command Data (7th Prior Ever

| | | | | |
|---|---|---|---|---|
| Active Head Restraint, Time to Deploy, Driver (msec) | SNA | Complete | 415.0 | Complete |
| Active Head Restraint, Time to Deploy, Front Passenger (msec) | SNA | 0 | 415.0 | 0 |
| Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver ( | 2.0 | 0 | SNA | 0 |
| Frontal Airbag Deployment, Time to 1st Stage Deployment, Front P | No | None | SNA | |
| Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec) | 12.0 | | 415.0 | 31901), (B1906), |
| Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (m | No | | 415.0 | B1836 1821) |
| Pretensioner Deployment, Time to Fire, 1st Seat, Driver (msec) | 1.5 | | | |
| Pretensioner Deployment, Time to Fire, 1st Seat, Passenger (msec | 1.5 | | | |
| Pretensioner Deployment, Time to Fire, 2nd Seat, Driver (msec) | SNA | | | |
| Pretensioner Deployment, Time to Fire, 2nd Seat, Passenger (msec | SNA | | | |
| Rear Window Airbag Deployment, Time to Deploy (msec) | SNA | | | |
| Side Airbag Deployment, Time to Deploy, 1st Seat, Driver (msec) | No | | | |
| Side Airbag Deployment, Time to Deploy, 1st Seat, Passenger (mse | 140.0 | | | |
| Side Airbag Deployment, Time to Deploy, 2nd Seat, Driver (msec) | SNA | | | |
| Side Airbag Deployment, Time to Deploy, 2nd Seat, Passenger (ms | SNA | | | |
| Side Curtain Airbag Deployment, Time to Deploy, Driver (msec) | 2.0 | | | |
| Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec | 2.0 | | | |

No DTC's yet

3 DTC's

Many DTC's

## DTCs Present at Time of Event (7th Prior Ev

| | | | | | |
|---|---|---|---|---|---|
| Recording Status, Diagnostic | Complete | Complete | 10 | 12 | 13 |
| Ignition Cycle Since DTC was Set (times) | 0 | 0 | Side Crash | Rollover | Side Crash |
| Airbag Warning Lamp ON Time Since DTC was Set (min) | 0 | (B1862), (B1837), (B1832) | Frontal/Rear/Side Crash | Rollover | No Event |
| Diagnostic Trouble Codes | None | | 1,441.0 | 2,592.0 | 32767 or greater |
| | | | OFF | OFF | OFF |
| | | | None | None | None |
| | | | Complete | Complete | Invalid |
| | | | 45,240 [72,807] | 45,240 [72,807] | 4434 |
| | | | 4431 | 4431 | 115,900 |
| | | | 1,715,800 | 1,718,600 | IG |
| | | | Normal | Normal | |

No DTC's yet

DTC's take
2-3 sec to set

# New **Look** to Precrash Data (same data)

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4**

| Time (sec) | Vehicle Speed (MPH [km/h]) | Accelerator Pedal, % Full (%) | Percentage of Engine Throttle (%) | Fuel Injection Quantity (mm^3/st) | Engine RPM (RPM) | Motor RPM (RPM) | Service Brake, ON/OFF |
|---|---|---|---|---|---|---|---|
| -5.00 | 29.8 [48] | 49.5 | 24.5 | Invalid | 2,400 | Invalid | OFF |
| -4.50 | 32.3 [52] | 47.0 | 24.5 | Invalid | 2,500 | Invalid | OFF |
| -4.00 | 34.2 [55] | 38.0 | 20.0 | Invalid | 2,700 | Invalid | OFF |
| -3.50 | 35.4 [57] | 32.5 | 16.0 | Invalid | 2,800 | Invalid | OFF |
| -3.00 | 36.7 [59] | 26.5 | 13.5 | Invalid | 2,400 | Invalid | OFF |
| -2.50 | 37.3 [60] | 22.5 | 11.0 | Invalid | 2,400 | Invalid | OFF |
| -2.00 | 37.3 [60] | 19.0 | 8.5 | Invalid | 2,200 | Invalid | OFF |
| -1.50 | 38.5 [62] | 19.0 | 6.0 | Invalid | 1,900 | Invalid | OFF |
| -1.00 | 37.9 [61] | 18.0 | 5.5 | Invalid | 1,900 | Invalid | OFF |
| -0.50 | 37.9 [61] | 18.0 | 5.0 | Invalid | 1,800 | Invalid | OFF |
| TRG(0) | 36.0 [58] | 100.0 | 26.0 | Invalid | 1,700 | Invalid | OFF |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4**

| Time (sec) | ABS Control Status | BOS Control Status | Brake Oil Pressure (Mpa) | Longitudinal Acceleration , VSC Sensor (m/s^2) | Yaw Rate (deg/s) | Steering Input (degrees) | Shift Position |
|---|---|---|---|---|---|---|---|
| -5.00 | OFF | OFF | 0.00 | 1.579 | -0.98 | -4.5 | D |
| -4.50 | OFF | OFF | 0.00 | 1.795 | 0.00 | -1.5 | D |
| -4.00 | OFF | OFF | 0.00 | 1.292 | 0.49 | 0.0 | D |
| -3.50 | OFF | OFF | 0.00 | 0.861 | 0.00 | -1.5 | D |
| -3.00 | OFF | OFF | 0.00 | 0.718 | 0.00 | -1.5 | D |
| -2.50 | OFF | OFF | 0.00 | -0.072 | 0.00 | -1.5 | D |
| -2.00 | OFF | OFF | 0.00 | 0.000 | 0.00 | -1.5 | D |
| -1.50 | OFF | OFF | 0.00 | 0.000 | -0.49 | -1.5 | D |
| -1.00 | OFF | OFF | 0.00 | 0.287 | 0.00 | -1.5 | D |
| -0.50 | OFF | OFF | 0.00 | -0.144 | 0.00 | -1.5 | D |
| TRG(0) | OFF | OFF | 0.48 | -1.507 | 61.00 | -19.5 | D |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4**

| Time (sec) | Sequential Shift Range | Cruise Control Status | VSC Control Status | READY Signal | Drive Mode, Power Train | Drive Mode, Snow | Drive Mode, EV |
|---|---|---|---|---|---|---|---|
| -5.00 | Undetermined | OFF | ON (enable) | Invalid | Normal | OFF | Invalid |

# 17 EDR time data & ped events

## System Status at Pedestrian Event (1st Prior Event, TRG 1)

| | |
|---|---|
| TRG Count (times) | 1 |
| TRG Count not for Pedestrian (times) | SNA |
| Ignition Cycle, Crash (times) Not present in side events | 353 |
| Odometer signal (miles [km]) | 1,195 [1,923] |
| Trip count (times) Similar to Ignition Cycles but slightly higher | 392 |
| Time count (msec) Ignition Time on to event | 1,283,900 |
| Time count input system | Normal |
| Airbag Warning Lamp, On/Off | OFF |
| Pedestrian Warning Lamp, On/Off | OFF |
| Time from Previous TRG (msec) | 32767 or greater |
| Time from Pre-Crash to TRG (msec) | 300 |
| Time between TRG not for Pedestrian and TRG for Pedestrian (msec) | SNA |
| TRG Establishment not for Pedestrian, within 5 sec. | ON |
| Freeze Signal | ON |
| Recording Status, Pre-Crash | Complete |
| Recording Status, Crash Info. | Complete |
| Recording Status, EDR Correlation Info. | Complete |

## Deployment Command Data (1st Prior Event, TRG 1)

| | |
|---|---|
| Front Pop-Up Hood, Time to Deploy (msec) | 3.0 |
| Rear Pop-Up Hood, Time to Deploy (msec) | 3.0 |
| Airbag for Pedestrian Deployment, Time to Deploy (msec) | SNA |

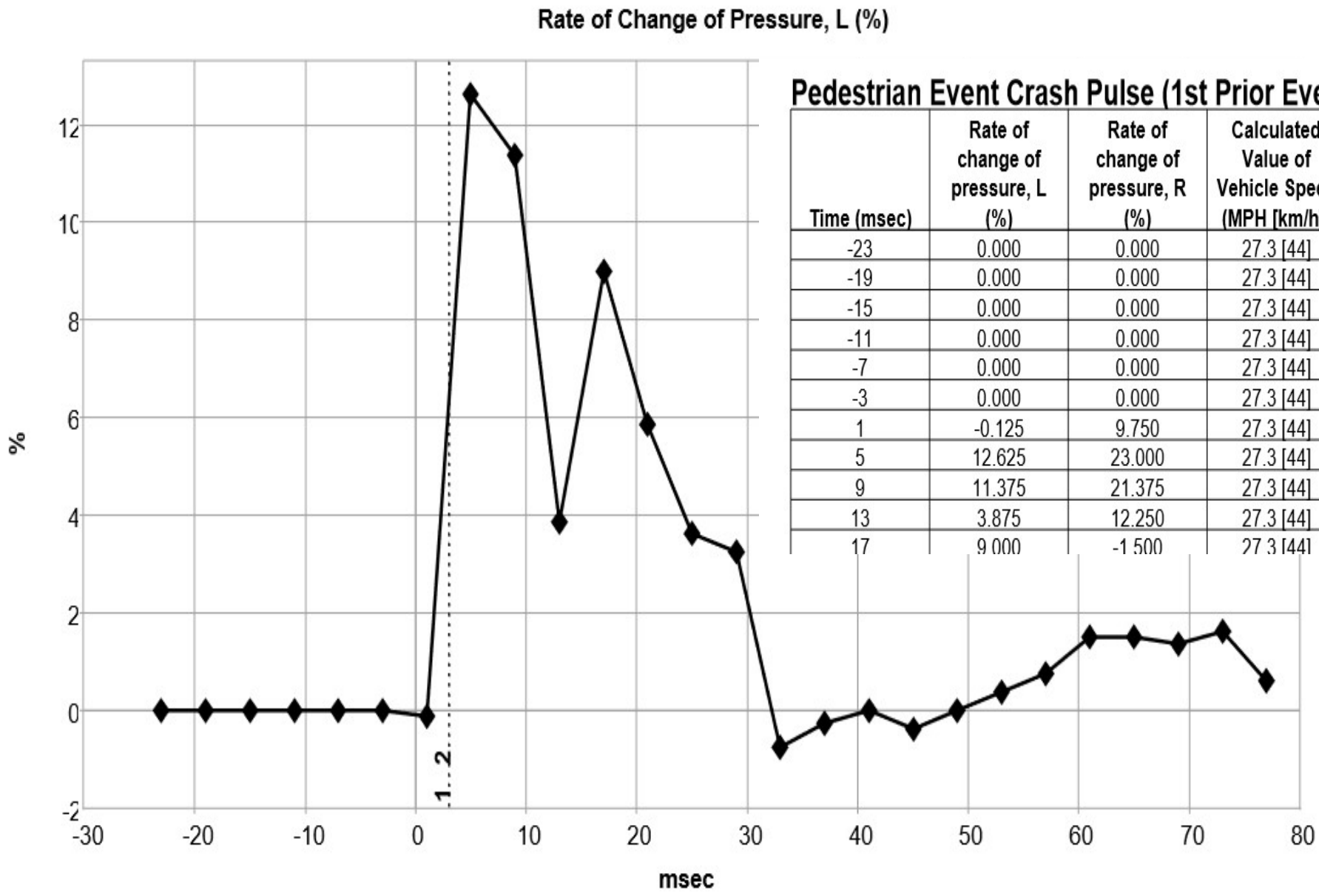
## DTCs Present at Time of Event (1st Prior Event, TRG 1)

| | |
|---|---|
| Recording Status, Diagnostic | Complete |
| Ignition Cycle Since DTC was Set (times) | 0 |
| Warning ON Time Since DTC was Set, for Pedestrian (min) | 0 |
| Diagnostic Trouble Codes | None |

## Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Time from TRG to Next Sample (msec) | 1 |

# Data from L/R bumper pressure sensors




**Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)**

| Time (msec) | Rate of change of pressure, L (%) | Rate of change of pressure, R (%) | Calculated Value of Vehicle Speed (MPH [km/h]) |
|---|---|---|---|
| -23 | 0.000 | 0.000 | 27.3 [44] |
| -19 | 0.000 | 0.000 | 27.3 [44] |
| -15 | 0.000 | 0.000 | 27.3 [44] |
| -11 | 0.000 | 0.000 | 27.3 [44] |
| -7 | 0.000 | 0.000 | 27.3 [44] |
| -3 | 0.000 | 0.000 | 27.3 [44] |
| 1 | -0.125 | 9.750 | 27.3 [44] |
| 5 | 12.625 | 23.000 | 27.3 [44] |
| 9 | 11.375 | 21.375 | 27.3 [44] |
| 13 | 3.875 | 12.250 | 27.3 [44] |
| 17 | 9.000 | -1.500 | 27.3 [44] |

# VCH Freeze Frame Data

- New feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, "**V**ehicle **C**ontrol **H**istory"
- Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion. 26 triggers including crash avoidance, ESC.
- Accessible only with Toyota Techstream service tool – not part of Bosch CDR system. Annual s/w subscription or 2 day subs available.
- CDR900 or Nexiq works as pass thru device same as Toyota Mongoose hardware
- Tabular Data exports as ".csv" –
- Space in memory for MANY events – can show a history of erratic driver behavior
- Details in SAE 2016-01-1495 Appendix D. New **2019-01-0632** "**Reconstructing Vehicle Dynamics from On-Board Event Data**" **discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events).**

# VCH Freeze Frame Data

- Early VCH systems have 26 different triggers and record only tabular EDR-style precrash data
- TSS-P & TSS-C, aka “1.0” systems, can record pictures if and only if the automatic emergency braking system engages (applies the brakes for you). A warning is not enough to record pictures.
- TSS “2.0” and “2.5” systems also record pictures in the event of a crash. See application chart that follows (best info available, not guaranteed).

# TOYOTA/LEXUS Vehicles with Vehicle Control History Data

| | | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LEXUS | ES | | | | | | | | LSS+ 2.0 | LSS+ 2.0 | LSS+ 2.5 | LSS+ 2.5 |
| | GS | | | | | | | | LSS+ | | | |
| | GX | | | | | | | | | | LSS+ | LSS+ |
| | IS | | 250 and 350 | | | | | | | LSS+ 2.5 | LSS+ 2.5 | LSS+ 2.5 |
| | LC | | | | | | | | | | LSS+ | LSS+ |
| | LS | | | | | | | | LSS+ 2.0 | LSS+ 2.0 | LSS+ 2.5 | LSS+ 2.5 & LSS+ A |
| | LX | | | | | | | | | | LSS+ 2.5 | LSS+ 2.5 |
| | NX | | | | | | | | | LSS+ 2.0 | LSS+ 3.0 | LSS+ 3.0 |
| | RC | | | | | | | | | | LSS+ | LSS+ 2.5 |
| | RX | | | | | | | | | LSS+ 2.0 | LSS+ 2.0 | LSS+ 3.0 |
| | RZ | | | | | | | | | | | LSS+ 3.0 |
| | UX | | | | | | | | LSS 2.0 | LSS 2.0 | LSS 2.0 | LSS+ 2.5 |
| TOYOTA | 4Runner | | | | ? | ? | | | | | TSS-P | TSS-P |
| | Avalon | | | | TSS-P | | | | | TSS-P | TSS 2.5+ | |
| | bZ4X | | | | | | | | | | | TSS 3.0 |
| | C-HR | | | | | | | | TSS-P | TSS 2.5 | TSS 2.5 | |
| | Camry | | | | | | | | TSS-P | TSS 2.5+ | TSS 2.5+ | TSS 2.5+ |
| | Corolla | | | | | | TSS-P | TSS 2.0 | TSS 2.0 | TSS 2.0 | TSS 2.0 | TSS 3.0 |
| | Crown | | | | | | | | | | | TSS 3.0 |
| | Highlander | | | | | | | TSS-P | TSS 2.0 | TSS 2.5+ | TSS 2.5+ | TSS 2.5+ |
| | Land Cruiser | | | | TSS-P | | | | | TSS-P | | |
| | Mirai | | | | | | | TSS 2.0 | TSS 2.0 | TSS 2.5+ | TSS 2.5+ | TSS 2.5+ |
| | Prius | | | | TSS-P | | | | TSS-P | TSS 2.0 | TSS 2.0 | TSS 3.0 |
| | Prius C | | | TSS-C | TSS-C | TSS-C | | | | | | |
| | Prius V | | | TSS-C | | | | | | | | |
| | Prius Prime | | | | | | | | | | | TSS 3.0 |
| | RAV4 | | | | TSS-P | | TSS-P | TSS 2.0 | TSS 2.0 | TSS 2.0 | TSS 2.0 | TSS 2.5 |
| | RAV4 Prime | | | | | | | | | | | TSS 2.5+ |
| | Sequoia | | | | | | | | | | TSS-P | TSS 2.5 |
| | Sienna | | | | Maybe? | Optional | | | TSS-P | TSS 2.0 | TSS 2.0 | TSS 2.0 |
| | Tacoma | | | | | | | | | | TSS-P | TSS-P |
| | Tundra | | | | | | | | | TSS-P | TSS 2.5 | TSS 2.5 |
| | Venza | | | | | | | | | TSS 2.0 | TSS 2.0 | TSS 2.5+ |
| | Yaris | | | | | TSS-C | Low Speed PCS | | | | | |
| SCION | iM | | | | | | | | | | | |

# Toyota/Lexus with Photos stored

Gen 1 TSS/LSS pictures only with auto braking, Gen 2 pictures in more circumsta

| | | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|---|---|
| LEXUS | CT200h | | | | discontinued | | | | | |
| | ES | | LSS | LSS | LSS | ? | ? | ? | 2.5 | |
| | GS | | LSS | LSS | LSS | ? | ? | discontinued | | |
| | GX | | | | | | LSS | LSS | LSS | |
| | IS | | | LSS | LSS | ? | ? | ? | 2.5 | |
| | LC | | | | LSS | LSS | LSS | LSS | LSS | |
| | LS | | | | LSS+ | 2.0 | 2.0 | 2.0 | 2.5 | |
| | LX | | LSS | LSS | LSS | ? | ? | ? | 2.5 | |
| | NX | | | | LSS | ? | ? | ? | 3.0 | |
| | RC-F | | | | LSS | ? | ? | ? | LSS | |
| | RX | | LSS | LSS | LSS | ? | ? | ? | 2.0 | |
| | UX | | | | | | | | 2.0 | |
| TOYOTA | 4Runner | | | | | | P | P | P | |
| | 86 | | | | | | | | Subaru | |
| | Avalon | | P | P | P | P | P | P | 2.5+ | |
| | Camry | | | | P | P | P | 2.5+ | 2.5+ | |
| | C-HR | | | | P | P | P | 2.5 | 2.5 | |
| | Corolla | | | P/C | P | 2.0 | 2.0 | 2.0 | 2.0 | |
| | Echo | | | | | | | | | |
| | FJ | | | | | | | | | |
| | Highlander | | | P | P | P | 2.0 | 2.5+ | 2.5+ | |
| | Land Cruiser | | P | P | P | P | P | P | | |
| | Mirai | | | | | 2.0 | 2.0 | 2.5+ | 2.5+ | |
| | Prius | | P/C | P/C | P | P/C | P | 2.0 | 2.0 | |
| | RAV4 | | P/C | P/C | P | 2.0 | 2.0 | 2.0 | 2.0 | |
| | Sequoia | | | | P | P/C | P | P | P | |
| | Sienna | | | | P | P/C | P | 2.0 | 2.0 | |
| | Supra (BMW) | | | | | | (made by BMW) | | | |
| | Tacoma | | | | P | P | P | P | P | |
| | Tundra | | | | P | P | P | P | 2.5 | |
| | Venza | out of production | | | | | | 2.0 | 2.0 | |
| | Yaris | | | C | C | | | | | |

Chart is only inteded to cover US models. Content MAY depend on trim level or model.

How to get the data

# Toyota TechStream / PCS & VCH Data

For IATAI 2021 Conference

This is only the lead slide of a 200+stand-alone slide presentation on how to extract the data. Review the complete presentation in your student reference materials to know how to get the data out.


**This presentation developed by**

**Brad Muir**

**of**

CRASH DATA SPECIALISTS

WWW.CDR-TRAINERS.COM

**Edited for time and Presented by**

**Rick Ruth**

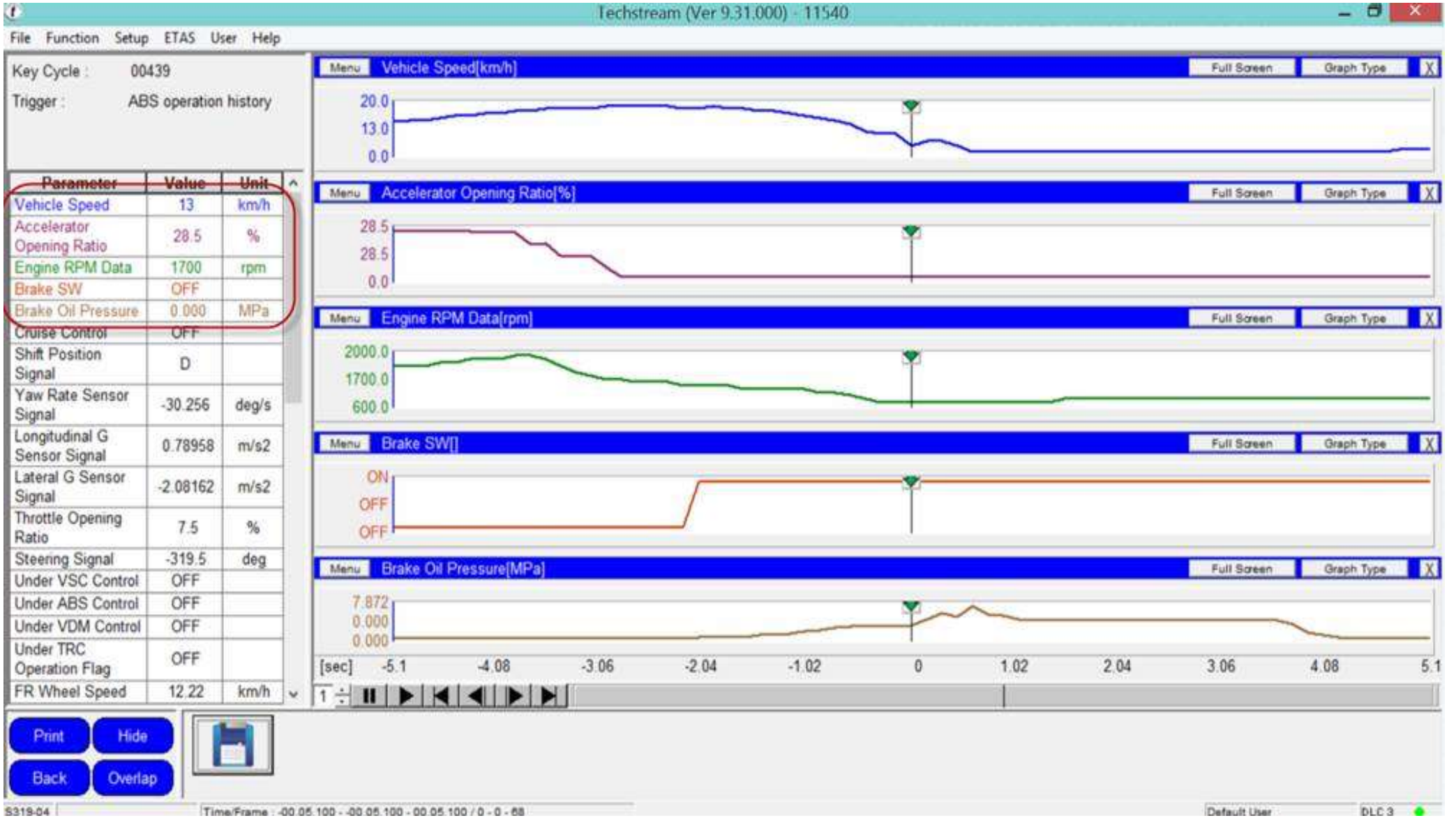
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IPTM

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# Detailed VCH Data Per Event



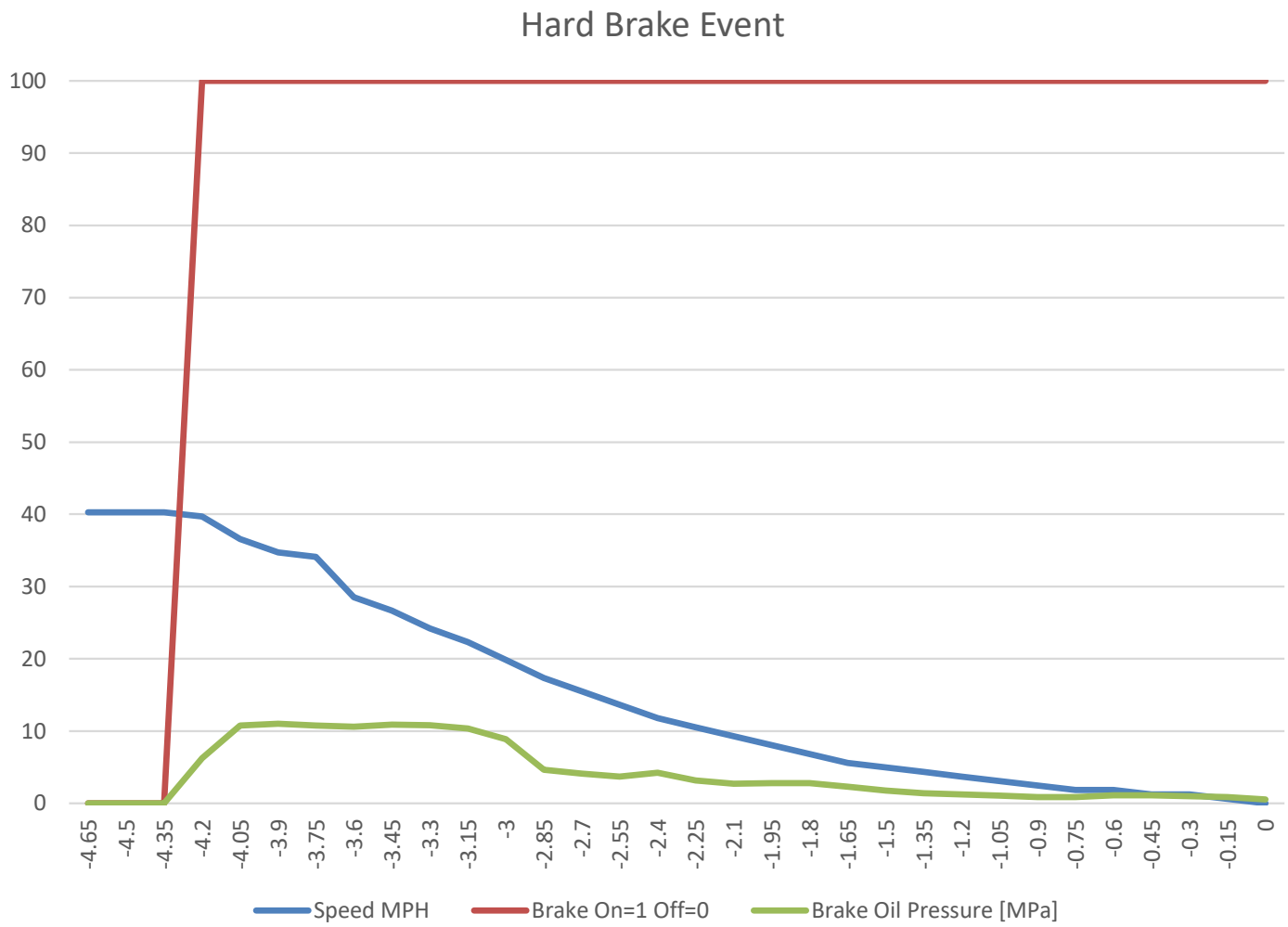
# Images from the Toyota FCM



# Event Data from Sudden Braking Event

| Date/Time Stamp | Time relative to stop | Vehicle Speed [km/h] | Speed MPH | Accelerator Opening Ratio [%] | Engine RPM Data [rpm] | Cruise Control | Shift Position Signal | Brake SW | Brake On=1 Off=0 | Brake Oil Pressure [MPa] | Yaw Rate Sensor Signal [deg/s] | Longitudinal G Sensor Signal | Long g's | Lateral G Sensor Signal [m/s^2] | Lat g's | Throttle Opening Ratio [%] | Steering Signal [deg] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2021-11-11-20:05:49.650 | -4.65 | 65 | 40.3 | 16.5 | 1200 | 0 | 1 | OFF | 0 | 0 | -2.928 | -0.14356 | -0.01462 | -0.57424 | -0.0586 | 8.5 | -4.5 |
| 2021-11-11-20:05:49.800 | -4.5 | 65 | 40.3 | 17.5 | 1200 | 0 | 1 | OFF | 0 | 0 | -2.928 | -0.14356 | -0.01462 | -0.3589 | -0.03662 | 6 | -4.5 |
| 2021-11-11-20:05:49.950 | -4.35 | 65 | 40.3 | 0 | 1200 | 0 | 1 | OFF | 0 | 0 | -3.416 | -0.14356 | -0.01462 | -0.50246 | -0.05127 | 4.5 | -4.5 |
| 2021-11-11-20:05:50.100 | -4.2 | 64 | 39.68 | 0 | 1200 | 0 | 1 | ON | 100 | 6.24 | -3.416 | -6.89088 | -0.70172 | -0.50246 | -0.05127 | 0 | -9 |
| 2021-11-11-20:05:50.200 | -4.05 | 59 | 36.58 | 0 | 1100 | 0 | 1 | ON | 100 | 10.752 | -4.392 | -8.9725 | -0.9137 | -0.50246 | -0.05127 | 0 | -18 |
| 2021-11-11-20:05:50.250 | -3.9 | 56 | 34.72 | 0 | 1100 | 0 | 1 | ON | 100 | 11.04 | -3.904 | -8.75716 | -0.89177 | -0.93314 | -0.09522 | 0 | -22.5 |
| 2021-11-11-20:05:50.400 | -3.75 | 55 | 34.1 | 0 | 1200 | 0 | 1 | ON | 100 | 10.752 | -6.832 | -8.9725 | -0.9137 | -0.93314 | -0.09522 | 0 | -27 |
| 2021-11-11-20:05:50.550 | -3.6 | 46 | 28.52 | 0 | 1000 | 0 | 1 | ON | 100 | 10.608 | -7.808 | -8.75716 | -0.89177 | -1.4356 | -0.14649 | 0 | -40.5 |
| 2021-11-11-20:05:50.700 | -3.45 | 43 | 26.66 | 0 | 900 | 0 | 1 | ON | 100 | 10.896 | -9.272 | -8.9725 | -0.9137 | -2.29696 | -0.23438 | 0 | -58.5 |
| 2021-11-11-20:05:50.850 | -3.3 | 39 | 24.18 | 0 | 800 | 0 | 1 | ON | 100 | 10.8 | -11.224 | -8.9725 | -0.9137 | -3.01476 | -0.30763 | 0 | -63 |
| 2021-11-11-20:05:51.000 | -3.15 | 36 | 22.32 | 0 | 800 | 0 | 1 | ON | 100 | 10.368 | -12.688 | -8.75716 | -0.89177 | -2.5123 | -0.25636 | 0 | -67.5 |
| 2021-11-11-20:05:51.150 | -3 | 32 | 19.84 | 0 | 900 | 0 | 1 | ON | 100 | 8.88 | -14.152 | -8.9725 | -0.9137 | -1.57916 | -0.16114 | 0 | -67.5 |
| 2021-11-11-20:05:51.300 | -2.85 | 28 | 17.36 | 0 | 800 | 0 | 1 | ON | 100 | 4.656 | -13.664 | -8.32648 | -0.84791 | -1.4356 | -0.14649 | 0 | -67.5 |
| 2021-11-11-20:05:51.450 | -2.7 | 25 | 15.5 | 0 | 800 | 0 | 1 | ON | 100 | 4.128 | -11.712 | -5.95774 | -0.60669 | -0.64602 | -0.06592 | 0 | -63 |
| 2021-11-11-20:05:51.600 | -2.55 | 22 | 13.64 | 0 | 700 | 0 | 1 | ON | 100 | 3.696 | -8.784 | -4.59392 | -0.46781 | -0.14356 | -0.01465 | 0 | -58.5 |
| 2021-11-11-20:05:51.750 | -2.4 | 19 | 11.78 | 0 | 700 | 0 | 1 | ON | 100 | 4.224 | -7.32 | -5.09638 | -0.51898 | 0.14356 | 0.014649 | 0 | -54 |
| 2021-11-11-20:05:51.900 | -2.25 | 17 | 10.54 | 0 | 700 | 0 | 1 | ON | 100 | 3.168 | -6.344 | -4.09146 | -0.41665 | 0.3589 | 0.036622 | 0 | -49.5 |
| 2021-11-11-20:05:52.050 | -2.1 | 15 | 9.3 | 0 | 700 | 0 | 1 | ON | 100 | 2.736 | -4.88 | -3.87612 | -0.39472 | 0.64602 | 0.06592 | 0 | -36 |
| 2021-11-11-20:05:52.200 | -1.95 | 13 | 8.06 | 0 | 700 | 0 | 1 | ON | 100 | 2.784 | -2.928 | -3.66078 | -0.37279 | 0.86136 | 0.087894 | 0 | -18 |
| 2021-11-11-20:05:52.350 | -1.8 | 11 | 6.82 | 0 | 700 | 0 | 1 | ON | 100 | 2.784 | -1.464 | -3.66078 | -0.37279 | 0.78958 | 0.080569 | 0 | -4.5 |
| 2021-11-11-20:05:52.500 | -1.65 | 9 | 5.58 | 0 | 600 | 0 | 1 | ON | 100 | 2.304 | -0.488 | -3.08654 | -0.31431 | 0.50246 | 0.051271 | 0 | 4.5 |
| 2021-11-11-20:05:52.650 | -1.5 | 8 | 4.96 | 0 | 600 | 0 | 1 | ON | 100 | 1.776 | 0 | -2.58408 | -0.26314 | 0.64602 | 0.06592 | 0 | 18 |
| 2021-11-11-20:05:52.800 | -1.35 | 7 | 4.34 | 0 | 600 | 0 | 1 | ON | 100 | 1.392 | 0.488 | -1.93806 | -0.19736 | 0.78958 | 0.080569 | 0 | 31.5 |
| 2021-11-11-20:05:52.950 | -1.2 | 6 | 3.72 | 0 | 600 | 0 | 1 | ON | 100 | 1.248 | 0.976 | -1.7945 | -0.18274 | 0.64602 | 0.06592 | 0 | 40.5 |
| 2021-11-11-20:05:53.100 | -1.05 | 5 | 3.1 | 0 | 600 | 0 | 1 | ON | 100 | 1.056 | 0.976 | -1.65094 | -0.16812 | 0.43068 | 0.043947 | 0 | 54 |
| 2021-11-11-20:05:53.250 | -0.9 | 4 | 2.48 | 0 | 600 | 0 | 1 | ON | 100 | 0.864 | 1.464 | -1.29204 | -0.13157 | 0.50246 | 0.051271 | 0 | 63 |
| 2021-11-11-20:05:53.400 | -0.75 | 3 | 1.86 | 0 | 600 | 0 | 1 | ON | 100 | 0.864 | 0.976 | -1.29204 | -0.13157 | 0.57424 | 0.058596 | 0 | 76.5 |
| 2021-11-11-20:05:53.550 | -0.6 | 3 | 1.86 | 0 | 600 | 0 | 1 | ON | 100 | 1.104 | 1.464 | -1.36382 | -0.13888 | 0.3589 | 0.036622 | 0 | 85.5 |
| 2021-11-11-20:05:53.700 | -0.45 | 2 | 1.24 | 0 | 600 | 0 | 1 | ON | 100 | 1.104 | 0.488 | -1.50738 | -0.1535 | 0.21534 | 0.021973 | 0 | 90 |
| 2021-11-11-20:05:53.850 | -0.3 | 2 | 1.24 | 0 | 600 | 0 | 1 | ON | 100 | 1.008 | 0.488 | -1.22026 | -0.12426 | 0.43068 | 0.043947 | 0 | 90 |
| 2021-11-11-20:05:54.000 | -0.15 | 1 | 0.62 | 0 | 600 | 0 | 1 | ON | 100 | 0.864 | 0.488 | -1.0767 | -0.10964 | 0.28712 | 0.029298 | 0 | 94.5 |
| 2021-11-11-20:05:54.150 | 0 | 0 | 0 | 0 | 600 | 0 | 1 | ON | 100 | 0.528 | 0 | -0.57424 | | 0.3589 | 0.036622 | 0 | 94.5 |
| 2021-11-11-20:05:54.300 | 0 | 0 | | 0 | 600 | 0 | 1 | ON | 100 | 0.576 | 0 | -0.64602 | | 0.3589 | 0.036622 | 0 | 99 |

# Graph from Hard Brake Event Data



# Toyota EDR Accuracy

- Toyota stated up front in 2009 that its first prototype readout tool was not validated –but that didn't mean it wasn't right.
- Because early Toyota EDR's (like many other manufacturers) that lost power at impact may not make a new recording at impact, there were examples of when the Toyota EDR's did not match the accident facts.
- Some claimed the Toyota EDR was unreliable - based on a lack of understanding of the not-yet-written data limitations.

# Specific Toyota Validation

- Niehoff 2005: 3 barrier crashes: Delta V ranged -6.7% to +1.1%, only vehicle with speed data was -0.9% of reference instrument.
- Gabler 2008: 11 frontal/4 side barrier crashes: Delta V data ranged from -22% to +11%, avg -1.6%. Only one vehicle had speed data, it reported 35 mph in a 33.8 mph test or 3% high. Belt status was reported correctly in all 15 tests
- NHTSA 2011: 14 staged events, “No errors in measurement and storage or pre crash data” (within known accuracy and timing limitations). Vehicle speed within 1.5 mph, throttle voltage within 2.5%.

# Additional Toyota Validation

- Ishikawa & Takubo 2009: 14 J-NCAP crashes EDR Delta V is from -14% to +0.5% of reference instrumentation, averaging -7.3%. Speed data within 4%, avg. 2%
- Ruth 2012: SAE 2012-01-0999 -Speed data within 3%
- Brown 2012: SAE 2012-01-0996 & 0998 (Actual speed -2.7%)
- Wilkinson 2013: SAE 2013-01-1268 (Gen 1 Delta V 1mph low in low speed frontal collisions, 1mph high in rear collisions)
- Webster 2014: SAE 2014-01-0505 **Accuracy of Recorded Driver Inputs in Toyota Part 563 EDR** (Only addressed accuracy of brake and accelerator pedal position).
- SAE 2016-01-1494, 2017-01-1438 continue Wilkinson 2013 work into other generations. Gen 1-2 ACM Delta V may be under reported slightly in frontal collisions, and over reported slightly in rear collisions

# Additional Toyota Validation

**2019-01-0633 Performance of Event Data Recorders Found in Toyota Airbag Control Modules in High Severity Frontal Oblique Offset Crash Tests** by William Bortles, Kineticorp et al (finds Delta V is understated due to offset).

**2019-01-0635 Laboratory Experiments Using a 2007 Toyota Auris Event Data Recorder and Additional Data**

**from CAN Bus by Miguel Tabone et al, University of Malta (took electrical system out of car into lab, injected ABS wheel speed sensor signals to simulate moving vehicle as events were set by hammer blows)**

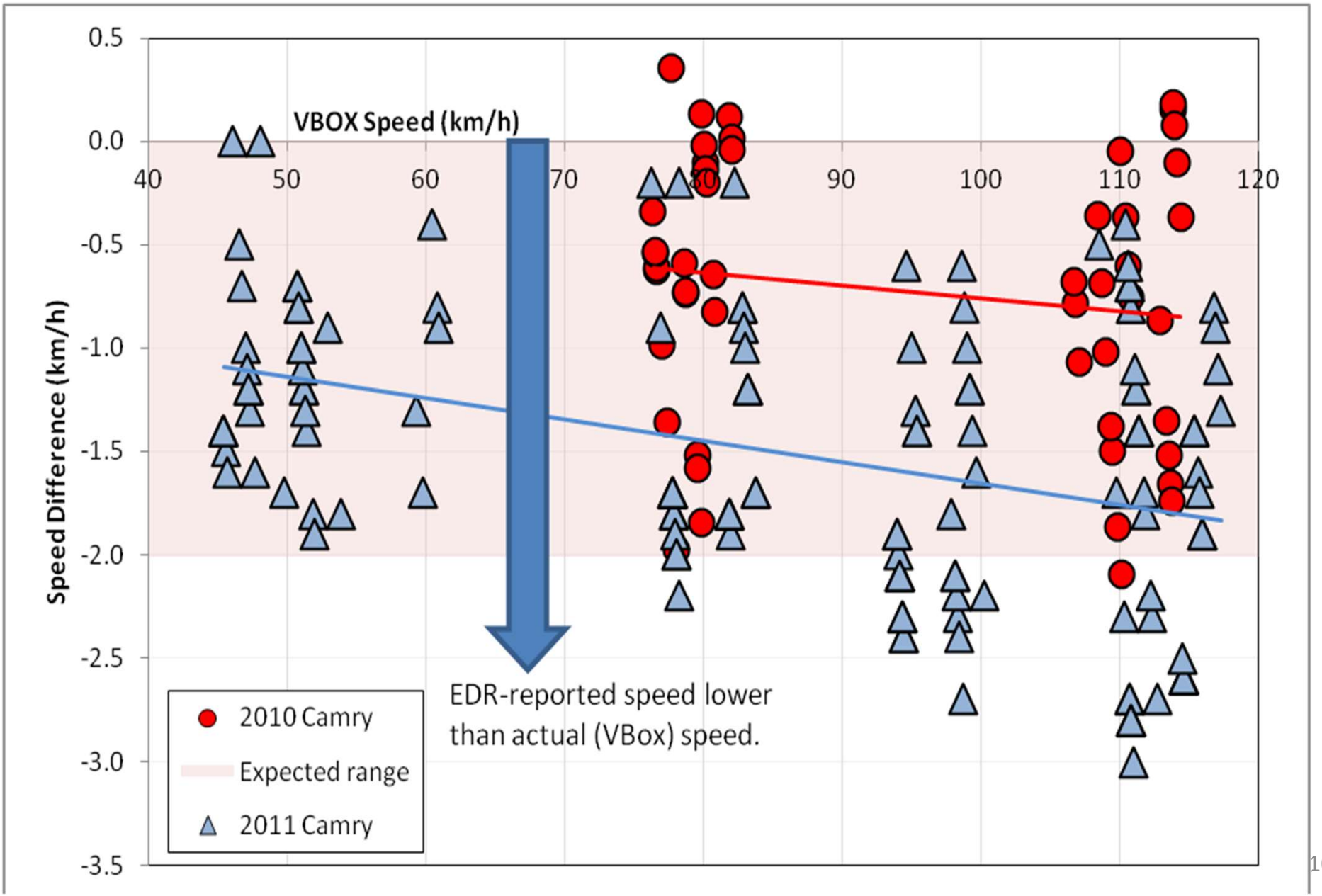
**2019-01-0632 Reconstructing Vehicle Dynamics from On-Board Event Data by Brandon Tsuge et al, MEA Forensic (read out Vehicle Control History, discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events)**

CONCLUSION

When the recording is from the event of interest, and data limitations are understood, there is no reason to believe speed data is any less accurate than any other manufacturer's EDR based on 42 test events. Delta V may under report slightly in Gen 1 and Gen 2 ACM's in frontal collisions and over reported in rear collisions, but in barrier tests are still within the generally accepted +/-10%. Other than the 00 EDR seat belt reporting at time of readout and not event (noted in the data limitations), there are no known issues with other data elements at this time.

# Toyota Speed Data Accuracy Validated

SAE 2012-01-0999  2010/11 Camry VBOX vs EDR



# The Accuracy and Sensitivity of 2005 to 2008 Toyota Corolla Event Data Recorders in Low-Speed Collisions
# SAE 2013-01-1268

- Induced a series of low speed events on vehicles and on a sled
- Verified 2G wakeup threshold thru testing
- Determined “offset” in Toyota accelerometers causing them to read 1.3 to 2.6 km/h low in frontal collisions and 0.6 to 2.2 km/h high in rear collisions.

# Toyota EDR Summary

- Toyota began installing EDR's in 2001MY; many Gen 1 and some Gen 2 have Delta V data only, others have DV and precrash data. Gen 1 & 2 data is similar to GM but has added precrash data point sampled at TRG.
- Unique Gen 1 &2 issues (75/78 mph max speed, accel pedal in volts, & last data point up to 0.5 seconds old) require special attention.
- Toyota EDR data is as ***reliable*** as other manufacturers within the data limitations.
- "12EDR" precrash has many additional data elements helpful to crash recon. But has no ACM Y Delta V in frontal events.
- "13EDR" adds synchronized Y accel data.
- New Vehicle Control History more frequent data, pics!!!
- Newest 17 EDR has synchronized X/Y Delta V